Sports Med (2016) 46:1419–1449
DOI 10.1007/s40279-016-0486-0

REVIEW ARTICLE

The Importance of Muscular Strength in Athletic Performance

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Published online: 2 February 2016

Abstract This review discusses previous literature that has examined the influence of muscular strength on various factors associated with athletic performance and the benefits of achieving greater muscular strength. Greater muscular strength is strongly associated with improved force-time characteristics that contribute to an athlete's overall performance. Much research supports the notion that greater muscular strength can enhance the ability to perform general sport skills such as jumping, sprinting, and change of direction tasks. Further research indicates that stronger athletes produce superior performances during sport specific tasks. Greater muscular strength allows an individual to potentiate earlier and to a greater extent, but also decreases the risk of injury. Sport scientists and practitioners may monitor an individual's strength characteristics using isometric, dynamic, and reactive strength tests and variables. Relative strength may be classified into strength deficit, strength association, or strength reserve phases. The phase an individual falls into may directly affect their level of performance or training emphasis. Based on the extant literature, it appears that there may be no substitute for greater muscular strength when it comes to improving an individual's performance across a wide range of both general and sport specific skills while simultaneously reducing their risk of injury when performing these skills. Therefore, sport scientists and practitioners should implement long-term training strategies that promote the greatest muscular strength within the required context of each sport/event. Future research should examine how force-time characteristics, general and specific sport skills, potentiation ability, and injury rates change as individuals transition from certain standards or the suggested phases of strength to another.

Key Points

This review discusses previous literature that examined the influence of muscular strength on various factors associated with athletic performance and the benefits of achieving greater muscular strength.

Greater muscular strength is associated with enhanced force-time characteristics (e.g. rate of force development and external mechanical power), general sport skill performance (e.g. jumping, sprinting, and change of direction), and specific sport skill performance, but is also associated with enhanced potentiation effects and decreased injury rates.

The extant literature suggests that greater muscular strength underpins many physical and performance attributes and can be vastly influential in improving an individual's overall performance.

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1 Introduction

A number of underlying factors may contribute to an athlete's performance. While sport scientists and practitioners cannot manipulate an athlete's genetic characteristics, an athlete's absolute and relative muscular strength can be enhanced with regular strength training. Muscular strength has been defined as the ability to exert force on an external object or resistance [1, 2]. Given the demands of an individual's sport or event, he or she may have to exert large forces against gravity in order to manipulate their own body mass (e.g., sprinting, gymnastics, diving, etc.), manipulate their own body mass plus an opponent's body mass (e.g., American football, rugby, wrestling, etc.), or manipulate an implement or projectile (e.g., baseball, weightlifting, shotput, etc.). The constant within all of the previous examples that may be considered a limiting factor of performance is the individual's muscular strength. The purpose of this review is to discuss previous literature that has examined the influence of muscular strength on various factors associated with athletic performance and to discuss the benefits of achieving greater muscular strength.

2 Literature Search Methodology

Original and review journal articles were retrieved from electronic searches of PubMed and Medline (EBSCO) databases. Additional searches of Google Scholar and relevant bibliographic hand searches with no limits of language of publication were also completed. The search strategy included the search terms 'maximum strength and jumping', 'maximum strength and sprinting', 'maximum strength and change of direction', 'maximum strength and power', 'strength and rate of force development', 'muscular strength and injury rate', 'strength level and postactivation potentiation', and 'strength level and athletic performance'. The last month of the search was January 2016.

The authors acknowledge that there are other methods of assessing muscular strength (e.g., isokinetic, dynamic strength index, etc.); however, this article focuses primarily on isometric and dynamic measures of strength. Furthermore, the authors acknowledge the disparity between lower and upper extremity strength literature as more scientific literature has examined lower extremity strength. This review uses a descriptive summary of research based on correlational analyses performed in each study. The magnitude of the relationships were defined as 0 to 0.3, or 0 to −0.3, was considered small; 0.31 to 0.49, or −0.31 to −0.49, moderate; 0.5 to 0.69, or −0.5 to −0.69, large; 0.7 to 0.89, or −0.7 to −0.89, very large; and 0.9 to 1.0, or −0.9 to −1.0, near perfect [3].

3 Influence of Strength on Force-Time Characteristics

High rates of force development (RFD) and subsequent high external mechanical power are considered to be two of the most important performance characteristics with regard to sport performance [4–6]. Previous research has indicated that RFD and power differs between starters and non-starters [7–12] and between different levels of athletes [8–11, 13–18]. Due to the importance of RFD and external mechanical power to an athlete's performance, trainable factors that may enhance these variables would be considered of utmost importance.

3.1 Rate of Force Development

Previous research has defined RFD as the rate of rise in force over the change in time, and has also been termed "explosive strength" [19]. The rate at which force can be produced is considered a primary factor to success in a large variety of sporting events [5]. The rationale behind this hypothesis is that a range of sports require the performance of rapid movements (e.g., jumping, sprinting, etc.) where there is a limited time to produce force (~50 to 250 ms) [20]. Similarly associated with force-time variables, impulse is defined as the product of force and the period of time in which the force is expressed. While impulse may ultimately determine vertical jump and weightlifting performance [21], the importance of RFD cannot be overlooked because a longer period of time (>300 ms) may be needed to reach maximum muscular force [19, 22–25]. Thus, the emphasis of training may be to increase RFD to allow a greater force to be produced over a given time period. This in turn would lead to an increase in the generated impulse or decrease in the time needed to obtain an equal impulse and subsequent acceleration of a person or implement.

Several studies have indicated that gaining strength through resistance training positively influences the RFD characteristics of an individual [19, 26–28]. Another study indicated that maximal muscular strength may account for as much as 80 % of the variance in voluntary RFD (150–250 ms) [20]. In support of these findings, a number of studies have examined the relationships between muscular strength and RFD (Table 1).

Fifty-nine Pearson correlation magnitudes were reported in Table 1 with all of the relationships being positive. Fifty-seven of the reported relationships (97 %) displayed a correlation magnitude of greater than or equal to 0.3, indicating a moderate relationship. Furthermore, 44 (75 %) of the reported correlation magnitudes displayed a large relationship with values of 0.5 or greater. Limited research

Table 1 Summary of studies correlating maximal strength and rate of force development variables

Study	Subjects (n)	Strength measure	RFD measure	Correlation results
Andersen et al. [26]	Healthy, sedentary males ($n = 15$)	MVC of knee extensors	Slope of torque–time curve in incrementing periods of 0–10, 0–20, 0–30,…0–250 ms	$r = 0.69$ (0–250 ms)
Bazyler et al. [29]	Recreationally-trained college-age males ($n = 17$)	1RM BS, 1RM partial squat, IS 90° knee angle, IS 120° knee angle	IRFD 0–250 ms with IS at 90° and 120°	IS at 90° IRFD: $r = 0.55$ (1RM BS), $r = 0.32$ (1RM partial squat), $r = 0.68$ (IS 90°), $r = 0.39$ (IS 120°) IS at 120° IRFD: $r = 0.43$ (1RM BS), $r = 0.42$ (1RM partial squat), $r = 0.45$ (IS 90°), $r = 0.64$ (IS 120°)
Beckham et al. [30]	Male and female intermediate to advanced weightlifters ($n = 12$)	IMTP	IRFD (0–100, 0–150, 0–200, 0–250 ms)	$r = 0.34$ (0–100 ms), $r = 0.42$ (0–150 ms), $r = 0.56$ (0–200 ms), $r = 0.73$ (0–200 ms)
Haff et al. [31]	Elite female weightlifters ($n = 6$)	1RM snatch and clean and jerk	Peak IRFD	$r = 0.79$ (Snatch) $r = 0.69$ (Clean and Jerk)
Kawamori et al. [32]	Collegiate male athletes ($n = 15$)	1RM HPC, Rel 1RM HPC	CMJ and SJ peak RFD	1RM HPC: $r = 0.53$ (CMJ), $r = 0.68$ (SJ) Rel 1RM HPC: $r = 0.56$ (CMJ), $r = 0.64$ (SJ)
Kawamori et al. [33]	Collegiate male weightlifters ($n = 8$)	IMTP	CMJ peak RFD, SJ peak RFD	$r = 0.85$ (CMJ), $r = 0.43$ (SJ)
Kraska et al. [34]	Male and female NCAA division I athletes ($n = 81$)	IMTP	IRFD (not specified)	$r = 0.88$
McGuigan et al. [35]	NCAA division III male wrestlers ($n = 8$)	IMTP	IRFD (not specified)	Not specified
McGuigan and Winchester [36]	NCAA division I male football players ($n = 22$)	1RM BS, IMTP	IRFD (not specified)	Not specified
Stone et al. [37]	International and local-level male cyclists ($n = 30$)	IMTP, Rel IMTP, IMTPa	Peak IRFD	IMTP: $r = 0.46$ Rel IMTP: $r = 0.23$ IMTPa: $r = 0.34$
Stone et al. [37]	National-level male and female cyclists ($n = 20$)	IMTP, Rel IMTP, IMTPa	Peak IRFD	IMTP: $r = 0.68$ Rel IMTP: $r = 0.20$ IMTPa: $r = 0.58$
Thomas et al. [38]	Male collegiate cricket, judo, rugby, and soccer athletes ($n = 22$)	Rel IMTP	Rel IRFD	$r = 0.70$

Table 1 continued

Study	Subjects (n)	Strength measure	RFD measure	Correlation results
Zaras et al. [39]	Young male and female track and field throwers (n = 12)	1RM BS, 1RM HPC, isometric leg press	Slope of force–time curve during 0–50, 0–100, 0–150, 0–200, and 0–250 ms	1RM BS and RFD 0–50 ms: r = 0.63 (Pre), r = 0.44 (Post); 0–100 ms: r = 0.70 (Pre), r = 0.65 (Post); 0–150 ms: r = 0.71 (Pre), r = 0.70 (Post); 0–200 ms: r = 0.59 (Pre), r = 0.69 (Post); 0–250 ms: r = 0.58 (Pre), r = 0.69 (Post) 1RM HPC and RFD 0–50 ms: r = 0.76 (Pre), r = 0.62 (Post); 0–100 ms: r = 0.77 (Pre), r = 0.77 (Post); 0–150 ms: r = 0.76 (Pre), r = 0.73 (Post); 0–200 ms: r = 0.64 (Pre), r = 0.71 (Post); 0–250 ms: r = 0.58 (Pre), r = 0.62 (Post) Isometric leg press and RFD 0–50 ms: r = 0.34 (Pre), r = 0.38 (Post); 0–100 ms: r = 0.59 (Pre), r = 0.72 (Post); 0–150 ms: r = 0.72 (Pre), r = 0.83 (Post); 0–200 ms: r = 0.73 (Pre), r = 0.86 (Post); 0–250 ms: r = 0.74 (Pre), r = 0.90 (Post)

1RM one repetition maximum, *BS* back squat, *CMJ* countermovement jump, *HPC* hang power clean, *IMTP* isometric mid-thigh clean pull, *IMTPa* allometrically-scaled isometric mid-thigh clean pull, *IRFD* isometric rate of force development, *IS* isometric squat, *MVC* maximal voluntary contraction, *Rel* relative, per kilogram of body mass, *RFD* rate of force development, *SJ* squat jump

has compared RFD values between stronger and weaker individuals. However, two of the previous studies indicated that stronger individuals produce greater RFD compared to those who are weaker [34, 38], while one study indicated that there was no statistical difference between the strongest and weakest individuals tested [37]. However, magnitude-based inferences from the latter study would indicate that there was a very large practical difference in RFD (Cohen's $d = 23.5$). Possible explanations for the lack of statistical differences between stronger and weaker groups in the latter study may be the small sample size in each group ($n = 6$) and the range in subject abilities within each group (e.g. Olympic training site cyclists and local cyclists).

3.2 External Mechanical Power

Previous research has indicated that external mechanical power may be the determining factor that differentiates the performance between athletes in sports [4, 40–51]. External mechanical power of the system reflects the sum of joint powers and may represent the coordinated effort of the lower body [52]. Therefore, instead of the sum of joint powers, system external mechanical power is often measured and has been related to a number of different sport performance characteristics such as sprinting [53, 54], jumping [55–58], change of direction [42, 59, 60], and throwing velocity [61, 62]. As a result, many have suggested that external mechanical power is one of the most important characteristics with regard to performance [4–6]. In fact, previous research has indicated that there were performance differences in external mechanical power between the playing level of athletes [10, 15, 18] and between starters and non-starters [7, 9–12]. As a result, it is not surprising that practitioners often seek to develop and improve external mechanical power in an effort to translate to improved sport performance.

Partly based on the concepts of Minetti [63] and Zamparo et al. [64], a periodization model has been developed termed phase potentiation [65, 66]. The idea behind this model is that the previous phase of training will potentiate or enhance the ability to realize specific physiological characteristics in a subsequent phase of training [67, 68]. For example, the completion of a strength-endurance phase, where the primary goals are to increase muscle cross-sectional area and work capacity, would enhance the ability to realize muscular strength characteristics in a maximal strength phase and a maximal strength phase would enhance the ability to realize muscular power characteristics in a subsequent strength-power or explosive speed phase of training. Taking the above into account, it would be logical that greater muscular strength would ultimately contribute to the ability to realize greater net joint power characteristics. A number of studies have indicated that the completion of strength training programs leads to an increase in absolute or relative external mechanical power [27, 51, 69–79]. The effectiveness of strength training programs may be explained by Newton's second law of motion (Σ forces acting on an object = object's mass • Object's acceleration). Within this law, the change in motion of an object (i.e. acceleration) is directly proportional to the forces impressed upon it. If greater forces are produced over a given period of time, a greater acceleration is produced, resulting in a greater velocity. Thus, increases in both force and velocity will ultimately result in an increase in power. Given that muscular strength has been defined as the ability to exert force on an external object or resistance [1, 2], practitioners must consider the importance of enhancing maximal strength when it comes to the development and improvement of external mechanical power. Previous research has examined the relationships between an individual's strength levels and external mechanical power (Table 2).

Collectively, the studies displayed in Table 2 reported 177 Pearson correlation coefficients. 134 of the reported correlation magnitudes (76 %) displayed moderate or greater relationship with strength, while 116 (65 %) displayed a correlation magnitude of greater than or equal to 0.5, indicating a large relationship. In support of these findings, several studies have examined external mechanical power performance differences between stronger and weaker subjects. Many of the studies indicated that the stronger subjects produced statistically greater external mechanical power characteristics as compared to their weaker counterparts [12, 14, 15, 37, 48, 55, 56, 85, 87–95], while only one study noted that no statistical differences existed between strong and weak subjects [38]. However, the authors of the latter article noted that the lack of statistical differences between strong and weak subjects may have been due to the lack of task homogeneity of the examined subjects which included cricket, judo, rugby, and soccer athletes. A second potential explanation may be the use of an isometric strength test compared to a dynamic strength test. The authors of the latter study indicated that dynamic strength tests may be more practical when assessing relationships between relative strength and dynamic performance. For a more detailed comparison, readers are directed to a review by Cormie and colleagues [96]. Taken collectively, the scientific literature suggests that muscular strength is highly correlated to external mechanical power and may be considered the foundation upon which external mechanical power can be built [25, 97, 98].

4 Influence of Strength on General Sport Skills

Some of the most common movements in sports are jumping, sprinting, and rapid change of direction (COD) tasks. The ability to perform these movements effectively may ultimately determine the outcome of certain events. As discussed previously, muscular strength can have a significant influence on important force-time characteristics related to performance. In theory, enhanced force-time characteristics should transfer to the ability to perform general sport skills. Therefore, the influence of muscular strength on jumping, sprinting, and COD cannot be overlooked.

4.1 Jumping

Jumping tasks, whether they are vertical or horizontal, are regularly performed and are often part of a larger skill set needed to be successful in sport competitions. In some instances, the ability to jump higher or farther than another competitor will determine who wins the competition (e.g., high jump, long jump, triple jump), while the repetitive nature of jumping tasks in other sports does not determine the winner. In team sports, jumping tasks may be used during rebounding in basketball, spiking/blocking in volleyball, diving in baseball, etc. While impulse may ultimately determine the jumping performance of an individual [21], distinct force-time characteristics may determine the shape and magnitude of the impulse created [99, 100]. As noted above, greater muscular strength may modify the force-time characteristics of an individual. Specifically, increases in muscular strength achieved through resistance training can alter both peak performance variables as well as the shape of the force-time curve [77, 89, 101]. Further research has indicated that stronger individuals may possess distinct force-time curve characteristics compared to weaker individuals (e.g., unweighted phase duration, relative shape of the jump phases, net impulse forces) [55, 77, 99]. Specifically, stronger subjects produced a shorter unweighted phase [99] and greater forces in the area of the force-time curve corresponding to net impulse compared to weaker subjects [55]. Moreover, increases in maximal strength following 10 weeks of strength training produced positive force adaptations during the late eccentric/early concentric phase of jump squats [77]. In support of previous research, a number of other studies examined the relationships between maximal strength and jumping performance (Table 3).

Collectively, the studies displayed in Table 3 reported 116 Pearson correlation magnitudes. Ninety-one of the reported correlation magnitudes (78 %) displayed a moderate or greater relationship with strength. Furthermore, 69

Table 2 Summary of studies correlating maximal strength and peak power variables

Study	Subjects (n)	Strength measure	PP measure	Correlation results
Baker and Nance [49]	Professional male rugby league players ($n = 20$)	3RM BS, 3RM BP, 3RM HPC	JS, incline BP throw	JS: $r = 0.81$ (3RM BS), $r = 0.79$ (3RM HPC) Incline BP throw: $r = 0.89$ (3RM BP), $r = 0.55$ (3RM HPC)
Baker [15]	National Rugby League and city-league college-aged rugby league males ($n = 49$)	1RM BP	BP throw at PP load	$r = 0.82$ (All), $r = 0.58$ (National Rugby League players), $r = 0.85$ (City-league players)
Baker et al. [80]	Professional and semiprofessionals male rugby league players ($n = 31$)	1RM BP	BP throw at PP load	$r = 0.66$ (Professional), $r = 0.85$ (Semiprofessional)
Carlock et al. [44]	National-level male and female junior and senior weightlifters ($n = 64$)	1RM BS, 1RM snatch, 1RM CandJ, Rel 1RM BS, Rel 1RM snatch, Rel 1RM CandJ	CMJ and SJ PP, Rel PP	1RM BS and CMJ PP: $r = 0.91$ (Men), 0.82 (Women), 0.92 (All); Rel CMJ PP: $r = -0.17$ (Men), 0.23 (Women), 0.39 (All); SJ PP: $r = 0.91$ (Men), 0.82 (Women), 0.93 (All); Rel SJ PP: $r = 0.42$ (Men), 0.33 (Women), 0.42 (All) 1RM Snatch and CMJ PP: $r = 0.93$ (Men), 0.76 (Women), 0.93 (All); Rel CMJ PP: $r = -0.10$ (Men), 0.15 (Women), 0.47 (All); SJ PP: $r = 0.93$ (Men), 0.76 (Women), 0.92 (All); Rel SJ PP: $r = 0.23$ (Men), 0.28 (Women), 0.60 (All) 1RM C&J and CMJ PP: $r = 0.90$ (Men), 0.76 (Women), 0.91 (All); Rel CMJ PP: $r = -0.19$ (Men), 0.17 (Women), 0.45 (All); SJ PP: $r = 0.90$ (Men), 0.76 (Women), 0.90 (All); Rel SJ PP: $r = 0.34$ (Men), 0.26 (Women), 0.59 (All) Rel 1RM BS: $r = 0.29$ (CMJ PP), $r = 0.49$ (Rel CMJ PP), $r = 0.24$ (SJ PP), $r = 0.72$ (Rel SJ PP) Rel 1RM Snatch: $r = 0.25$ (CMJ PP), $r = 0.53$ (Rel CMJ PP), $r = 0.20$ (SJ PP), $r = 0.74$ (Rel SJ PP) Rel 1RM C&J: $r = 0.19$ (CMJ PP), $r = 0.71$ (Rel CMJ PP), $r = 0.14$ (SJ PP), $r = 0.71$ (Rel SJ PP)
Cronin and Hansen [81]	Professional male rugby league players ($n = 16$)	3RM BS	30 kg JS average power, Rel 30 kg JS average power	$r = 0.42$ (JS average power), $r = 0.15$ (Rel 30 kg JS average power)
Haff et al. [31]	Elite female weightlifters ($n = 6$)	IMTP	CMJ, SJ PP	$r = 0.88$ (CMJ) $r = 0.92$ (SJ)
Jones et al. [82]	Recreationally-trained males ($n = 29$)	1RM BS	CMJ PP, average power	$r = 0.70$ (PP), $r = 0.67$ (Average power)
Kawamori et al. [32]	Collegiate male athletes ($n = 15$)	1RM HPC, Rel 1RM HPC	CMJ and SJ PP	1RM HPC: $r = 0.68$ (CMJ), $r = 0.71$ (SJ) Rel 1RM HPC: $r = 0.45$ (CMJ), $r = 0.46$ (SJ)
Kawamori et al. [33]	Collegiate male weightlifters ($n = 8$)	IMTP	CMJ, SJ PP	$r = 0.95$ (CMJ), $r = 0.70$ (SJ)
Moss et al. [51]	Well-trained male physical education students ($n = 30$)	1RM Elbow flexion	Elbow flexion PP, elbow flexion with 2.5 kg PP	$r = 0.93$ (Elbow flexion PP), $r = 0.73$ (Elbow flexion with 2.5 kg)

Table 2 continued

Study	Subjects (n)	Strength measure	PP measure	Correlation results
Nuzzo et al. [46]	Male NCAA division I AA football and track and field athletes ($n = 12$)	1RM BS, 1RM PC, IS (140° knee angle), IMTP, Rel 1RM BS, Rel 1RM PC, Rel IS, Rel IMTP	CMJ PP, Rel CMJ PP	PP: $r = 0.84$ (1RM BS), $r = 0.86$ (1RM PC), $r = 0.71$ (IS), $r = 0.75$ (IMTP) Rel PP: $r = 0.68$ (Rel 1RM BS), $r = 0.71$ (Rel 1RM PC), $r = 0.27$ (Rel IS), $r = 0.51$ (Rel IMTP)
Peterson et al. [83]	First-year male and female collegiate athletes ($n = 55$)	1RM BS, Rel 1RM BS	CMJ PP	1RM BS: $r = 0.92$ (All), $r = 0.66$ (Males), $r = 0.72$ (Females) Rel 1RM BS: $r = 0.69$ (All), $r = 0.39$ (Males), $r = -0.06$ (Females)
Requena et al. [84]	Professional male soccer players ($n = 21$)	1RM COHS, knee extensor MVC, plantar flexor MVC	COHS PP with 50, 75, 100, and 125 % of BM loads	50 % BM PP: $r = 0.59$ (1RM COHS), $r = 0.60$ (Knee Extensor MVC), $r = 0.61$ (Plantar Flexor MVC) 75 % BM PP: $r = 0.66$ (1RM COHS), $r = 0.65$ (Knee Extensor MVC), $r = 0.49$ (Plantar Flexor MVC) 100 % BM PP: $r = 0.83$ (1RM COHS), $r = 0.67$ (Knee Extensor MVC), $r = 0.57$ (Plantar Flexor MVC) 125 % BM PP: $r = 0.75$ (1RM COHS), $r = 0.58$ (Knee Extensor MVC), $r = 0.39$ (Plantar Flexor MVC)
Sheppard et al. [85]	International-level male volleyball players ($n = 21$)	Rel 1RM BS, Rel 1RM PC	CMJ Rel PP, CMJ +50 % body mass Rel PP	CMJ Rel PP: $r = 0.52$ (Rel 1RM BS), $r = 0.50$ (Rel 1RM PC) CMJ +50 % body mass Rel PP: $r = 0.59$ (Rel 1RM BS), $r = 0.51$ (Rel 1RM PC)
Speranza et al. [86]	Male first grade ($n = 10$), second grade ($n = 12$), and under 20 s ($n = 14$) semi-professional rugby league players	3RM BS, Rel 3RM BS	CMJ PP	3RM BS: $r = 0.44$ (All), $r = 0.57$ (First grade), $r = 0.35$ (Second grade), $r = 0.36$ (Under 20 s) Rel 3RM BS: $r = 0.11$ (All), $r = 0.38$ (First grade), $r = 0.10$ (Second grade), $r = -0.02$ (Under 20 s)
Speranza et al. [86]	Male first grade ($n = 10$), second grade ($n = 12$), and under 20 s ($n = 14$) semi-professional rugby league players	3RM BP, Rel 3RM BP	Plyometric push-up PP	3RM BP: $r = 0.43$ (All), $r = 0.79$ (First grade), $r = 0.07$ (Second grade), $r = 0.55$ (Under 20 s) Rel 3RM BP: $r = 0.09$ (All), $r = 0.02$ (First grade), $r = 0.30$ (Second grade), $r = -0.18$ (Under 20 s)
Stone et al. [48]	Males with BS experience ranging 7 weeks – 15 + years ($n = 22$)	1RM BS	CMJ, SJ at 10–100 % 1RM BS	CMJ: $r = 0.78$ (10 %), $r = 0.84$ (20 %), $r = 0.85$ (30 %), $r = 0.88$ (40 %), $r = 0.88$ (50 %), $r = 0.85$ (60 %), $r = 0.84$ (70 %), $r = 0.80$ (80 %), $r = 0.73$ (90 %), $r = 0.60$ (100 %) SJ: $r = 0.84$ (10 %), $r = 0.87$ (20 %), $r = 0.90$ (30 %), $r = 0.94$ (40 %), $r = 0.94$ (50 %), $r = 0.93$ (60 %), $r = 0.90$ (70 %), $r = 0.91$ (80 %), $r = 0.86$ (90 %), $r = 0.75$ (100 %)
Stone et al. [37]	International and local-level male cyclists ($n = 30$)	IMTP, Rel IMTP, IMTPa	CMJ and SJ PP, Rel PP, PPa	IMTP: $r = 0.79$ (CMJ PP), $r = 0.49$ (CMJ Rel PP), $r = 0.67$ (CMJ PPa), $r = 0.78$ (SJ PP), $r = 0.42$ (SJ Rel PP), $r = 0.62$ (SJ PPa) Rel IMTP: $r = 0.40$ (CMJ PP), $r = 0.43$ (CMJ Rel PP), $r = 0.44$ (CMJ PPa), $r = 0.39$ (SJ PP), $r = 0.40$ (SJ Rel PP), $r = 0.42$ (SJ PPa) IMTPa: $r = 0.60$ (CMJ PP), $r = 0.48$ (CMJ Rel PP), $r = 0.57$ (CMJ PPa), $r = 0.59$ (SJ PP), $r = 0.43$ (SJ Rel PP), $r = 0.54$ (SJ PPa)

Table 2 continued

Study	Subjects (*n*)	Strength measure	PP measure	Correlation results
Stone et al. [37]	National-level male and female cyclists ($n = 20$)	IMTP, Rel IMTP, IMTPa	CMJ and SJ PP, Rel PP, PPa	IMTP: $r = 0.85$ (CMJ PP), $r = 0.68$ (CMJ Rel PP), $r = 0.78$ (CMJ PPa), $r = 0.86$ (SJ PP), $r = 0.69$ (SJ Rel PP), $r = 0.78$ (SJ PPa) Rel IMTP: $r = 0.41$ (CMJ PP), $r = 0.56$ (CMJ Rel PP), $r = 0.52$ (CMJ PPa), $r = 0.42$ (SJ PP), $r = 0.62$ (SJ Rel PP), $r = 0.56$ (SJ PPa) IMTPa: $r = 0.64$ (CMJ PP), $r = 0.68$ (CMJ Rel PP), $r = 0.73$ (CMJ PPa), $r = 0.61$ (SJ PP), $r = 0.72$ (SJ Rel PP), $r = 0.76$ (SJ PPa)
Thomas et al. [38]	Male collegiate cricket, judo, rugby, and soccer athletes ($n = 22$)	IMTP, Rel IMTP	CMJ, SJ PP	IMTP: $r = 0.34$ (CMJ), $r = 0.46$ (SJ) Rel IMTP: $r = 0.01$, $r = 0.15$ (SJ)

1RM one repetition maximum, *3RM* three repetition maximum, *BP* bench press, *BS* back squat, *C&J* clean and jerk, *CMJ* countermovement jump, *COHS* concentric-only half-squat, *HPC* hang power clean, *IMTP* isometric mid-thigh clean pull, *IMTPa* allometrically-scaled isometric mid-thigh clean pull, *IS* isometric squat, *JS* jump squat, *MVC* maximal voluntary contraction, *PC* power clean, *PP* peak power, *PPa* allometrically-scaled peak power, *Rel* relative, per kilogram of body mass, *SJ* squat jump

correlation magnitudes (59 %) were greater than or equal to 0.5, indicating a large relationship. In support of these findings, several studies indicated that stronger individuals jumped higher compared to weaker individuals [12, 34, 50, 56, 85]. In contrast, one study indicated that there was no difference in jump height between strong and weak subjects [38]. A potential explanation for the latter findings may include the lack of task homogeneity of the subjects and the use of an isometric strength test compared to a dynamic strength test to compare dynamic performance.

4.2 Sprinting

The ability to accelerate rapidly and reach high sprinting speeds is a vital component of many sports or events. While peak sprinting speeds may dictate the winner of certain track events (e.g., 100, 200 m, etc.), athletes playing field sports such as soccer, rugby, lacrosse, and field hockey may not necessarily reach their maximum velocity regularly [113]. In fact, the average sprint time in soccer [114] and rugby union [115] is approximately 2 s covering distances of about 14 m [116] and 20 m [117, 118], respectively. Further research indicated that rugby union players may only reach approximately 70 % of their maximum sprinting speed after sprinting for 2 s [119]. Thus, it would appear that the ability to accelerate over short distances may be paramount for field athletes.

Previous research indicated that elite athletes produced greater speeds over short distances compared to non-elite athletes [120]. Faster runners possess several characteristics such as greater force application, shorter ground contact times, and greater stride lengths [54]. Further research indicated that sprint performance may be limited by the ability to produce a high RFD over the brief contacts instead of the ability to apply force [53]. In fact, better sprinters are able to generate greater vertical forces within the first half of their stance phase [16]. As displayed above, maximal strength is strongly correlated with RFD and thus, it is logical that sprinting performance would also be related to the strength level of individuals. Previous research has indicated that increases in strength coincide with increases in short sprint performance [121–125]. In support of these findings, a number of studies have examined the relationships between maximal strength and sprinting performance (Table 4).

Better sprinting performances are indicated by faster sprint times and higher speeds. Collectively, 67 correlation magnitudes between strength and sprinting performance were reported in Table 4. Of those within the table, 57 reported a moderate or greater relationship with strength (85 %), while 44 (66 %) displayed substantial relationships with strength. The correlation results presented in Table 4 are supported by a recent meta-analysis that indicated that

Table 3 Summary of studies correlating maximal strength and jump height/distance

Study	Subjects (n)	Strength measure	Jump type	Correlation results
Augustsson and Thomeé [102]	Recreationally-trained males ($n = 16$)	3RM BS	CMJ	$r = 0.51$
Blackburn et al. [45]	Healthy female college students ($n = 20$)	1RM BS, 1RM knee extension	CMJ, broad jump	1RM BS: $r = 0.65$ (CMJ), $r = 0.72$ (Broad jump) 1RM Knee extension: $r = 0.10$ (CMJ), 0.07 (Broad jump)
Carlock et al. [44]	National-level male and female junior and senior weightlifters ($n = 64$)	1RM BS, 1RM snatch, 1RM CandJ, Rel 1RM BS, Rel 1RM snatch, Rel 1RM CandJ	CMJ, SJ	CMJ: $r = 0.52$ (1RM BS), $r = 0.60$ (1RM Snatch), $r = 0.59$ (1RM CandJ), $r = 0.69$ (Rel 1RM BS), $r = 0.76$ (Rel 1RM Snatch), $r = 0.72$ (Rel 1RM CandJ) SJ: $r = 0.58$ (1RM BS), $r = 0.64$ (1RM Snatch), $r = 0.64$ (1RM C&J), $r = 0.72$ (Rel 1RM BS), $r = 0.75$ (Rel 1RM Snatch), $r = 0.72$ (Rel 1RM C&J)
Comfort et al. [103]	Well-trained youth male soccer players ($n = 34$)	1RM BS, Rel 1RM BS	CMJ, SJ	1RM BS: $r = 0.76$ (CMJ), $r = 0.76$ (SJ) Rel 1RM BS: $r = 0.62$ (CMJ), $r = 0.64$ (SJ)
Cronin and Hansen [81]	Professional male rugby league players ($n = 16$)	3RM BS	CMJ, 30 kg JS	$r = 0.14$ (CMJ), $r = 0.16$ (30 kg JS)
Jones et al. [82]	Recreationally-trained males ($n = 29$)	1RM BS	CMJ, broad jump	$r = 0.22$ (CMJ), $r = 0.17$ (Broad jump)
Kawamori et al. [32]	Collegiate male athletes ($n = 15$)	1RM HPC, Rel 1RM HPC	CMJ, SJ	1RM HPC: $r = 0.13$ (CMJ), $r = 0.09$ (SJ) Rel 1RM HPC: $r = 0.56$ (CMJ), $r = 0.47$ (SJ)
Kawamori et al. [33]	Collegiate male weightlifters ($n = 8$)	IMTP	CMJ, SJ	$r = 0.82$ (CMJ), $r = 0.87$ (SJ)
Koch et al. [104]	Male and female track and field athletes ($n = 11$) Untrained male and female subjects ($n = 21$)	1RM BS	Broad jump	$r = 0.81$
Kraska et al. [34]	Male and female NCAA division I athletes ($n = 81$)	IMTP, IMTPa	CMJ, SJ LCMJ, LSJ	IMTP: $r = 0.36$ (CMJ) $r = 0.40$ (SJ), $r = 0.55$ (LCMJ), $r = 0.55$ (LSJ) IMTPa: $r = 0.41$ (CMJ) $r = 0.41$ (SJ), $r = 0.52$ (LCMJ), $r = 0.52$ (LSJ)
Loturco et al. [105]	Male and female Brazilian national team boxers ($n = 15$)	IS (90° knee angle)	CMJ, SJ	$r = 0.79$ (CMJ), $r = 0.79$ (SJ)
McGuigan et al. [35]	NCAA division III male wrestlers ($n = 8$)	IMTP	CMJ	$r =$ Not specified
McGuigan and Winchester [36]	NCAA division I male football players ($n = 22$)	1RM BS, IMTP	CMJ, broad jump	CMJ: $r = 0.54$ (1RM BS), Not specified for IMTP Broad jump: Not specified for 1RM BS or IMTP
McGuigan et al. [106]	Recreationally-trained males ($n = 26$)	1RM BS, IMTP, 1RM BP	CMJ	$r = 0.69$ (1RM BS), $r = 0.72$ (IMTP), $r = 0.70$ (1RM BP)

Table 3 continued

Study	Subjects (n)	Strength measure	Jump type	Correlation results
Nimphius et al. [42]	Female Australian Institute of Sport state softball players ($n = 10$)	Rel 1RM BS	CMJ	$r = 0.36$ (Pre-season), $r = 0.38$ (Mid-season), $r = 0.16$ (Post-season)
Nuzzo et al. [46]	Male NCAA division I AA football and track and field athletes ($n = 12$)	1RM BS, 1RM PC, IS (140° knee angle), IMTP, Rel 1RM BS, Rel 1RM PC, Rel IS, Rel IMTP	CMJ	$r = 0.22$ (1RM BS), $r = 0.06$ (1RM PC), $r = -0.07$ (IS), $r = 0.28$ (IMTP), $r = 0.69$ (Rel 1RM BS), $r = 0.64$ (Rel 1RM PC), $r = 0.28$ (Rel IS), $r = 0.59$ (Rel IMTP)
Peterson et al. [83]	First-year male and female collegiate athletes ($n = 55$)	1RM BS, Rel 1RM BS	CMJ, broad jump	1RM BS and CMJ: $r = 0.86$ (All), $r = 0.54$ (Males), $r = 0.37$ (Females); Broad jump: $r = 0.77$ (All), $r = 0.45$ (Males), $r = 0.31$ (Females) Rel 1RM BS and CMJ: $r = 0.85$ (All), $r = 0.67$ (Males), $r = 0.55$ (Females); Broad jump: $r = 0.81$ (All), $r = 0.53$ (Males), $r = 0.64$ (Females)
Requena et al. [84]	Professional male soccer players ($n = 21$)	1RM COHS, knee extensor MVC, plantar flexor MVC	CMJ, SJ	CMJ: $r = 0.50$ (1RM COHS), $r = 0.57$ (Knee Extensor MVC), $r = 0.14$ (Plantar Flexor MVC) SJ: $r = 0.50$ (1RM COHS), $r = 0.55$ (Knee Extensor MVC), $r = 0.30$ (Plantar Flexor MVC)
Secomb et al. [107]	International-level male surfers ($n = 15$)	IMTP	CMJ, SJ	$r = 0.65$ (CMJ), $r = 0.58$ (SJ)
Secomb et al. [108]	Junior male and female competitive surfers ($n = 30$)	IMTP, Rel IMTP	CMJ, SJ	IMTP: $r = 0.48$ (CMJ), $r = 0.48$ (SJ) Rel IMTP: $r = 0.46$, $r = 0.40$ (SJ)
Sheppard et al. [85]	International-level male volleyball players ($n = 21$)	Rel 1RM BS, Rel 1RM PC	CMJ, Rel CMJ, spike jump, Rel spike jump, DJ, Rel DJ	CMJ: $r = -0.44$ (Rel 1RM BS), $r = -0.40$ (Rel 1RM PC) Rel CMJ: $r = 0.54$ (Rel 1RM BS), $r = 0.53$ (Rel 1RM PC) Spike jump: $r = -0.06$ (Rel 1RM BS), $r = -0.01$ (Rel 1RM PC) Rel Spike jump: $r = 0.64$ (Rel 1RM BS), $r = 0.65$ (Rel 1RM PC) DJ: $r = -0.35$ (Rel 1RM BS), $r = -0.29$ (Rel 1RM PC) Rel DJ: $r = 0.55$ (Rel 1RM BS), $r = 0.55$ (Rel 1RM PC)
Stone et al. [37]	International and local-level male cyclists ($n = 30$)	IMTP, Rel IMTP, IMTPa	CMJ, SJ	IMTP: $r = 0.59$ (CMJ), $r = 0.51$ (SJ) Rel IMTP: $r = 0.45$ (CMJ), $r = 0.42$ (SJ) IMTPa: $r = 0.54$ (CMJ), $r = 0.48$ (SJ)
Stone et al. [37]	National-level male and female cyclists ($n = 20$)	IMTP, Rel IMTP, IMTPa	CMJ, SJ	IMTP: $r = 0.67$ (CMJ), $r = 0.66$ (SJ) Rel IMTP: $r = 0.59$ (CMJ), $r = 0.61$ (SJ) IMTPa: $r = 0.67$ (CMJ), $r = 0.68$ (SJ)
Thomas et al. [38]	Male collegiate cricket, judo, rugby, and soccer athletes ($n = 22$)	IMTP, Rel IMTP	CMJ, SJ	IMTP: $r = -0.02$ (CMJ), $r = -0.04$ (SJ) Rel IMTP: $r = -0.09$, $r = -0.10$ (SJ)
Ugarkovic et al. [109]	Junior male basketball players ($n = 33$)	Rel hip extensor MVC, Rel knee extensor MVC	CMJ	$r = 0.38$ (Rel hip extensor MVC), $r = 0.52$ (Rel knee extensor MVC)
Wisløff et al. [110]	Norwegion elite male soccer players ($n = 29$)	1RM BS	CMJ	$r = 0.61$

Table 3 continued

Study	Subjects (*n*)	Strength measure	Jump type	Correlation results
Wisløff et al. [50]	International male soccer players ($n = 17$)	1RM HS	CMJ	$r = 0.78$
Yamauchi and Ishii [111]	Untrained men and women ($n = 67$)	Isometric knee-hip extension peak force on servo-controlled dynamometer, Rel peak force	CMJ	$r = 0.48$ (Peak force), $r = 0.24$ (Rel Peak force)
Young et al. [112]	Males with at least 1 year of jumping experience ($n = 29$)	Rel IS (120° knee angle)	CMJ, run-up jump	$r = 0.33$ (CMJ), $r = 0.33$ (Run-up jump)

1RM one repetition maximum; *3RM* three repetition maximum; *BP* bench press; *BS* back squat; *C&J* clean and jerk; *CMJ* countermovement jump; *COHS* concentric-only half-squat; *DJ* depth, drop jump; *HPC* hang power clean; *HS* half-squat; *IMTP* isometric mid-thigh clean pull; *IMTPa* allometrically-scaled isometric mid-thigh clean pull; *IS* isometric squat; *JS* jump squat; *PC* power clean; *Rel* relative, per kilogram of body mass; *SJ* squat jump

increases in lower body strength positively transfer to sprinting performance [134]. Further research indicated that stronger individuals produced faster sprinting performances compared to those who were weaker [50, 56, 81, 130, 131, 135], while some research indicated that there was no difference between strong and weak subjects [14, 81]. A potential explanation for the conflicting findings was the use of only absolute strength measures in both investigations [14, 81] without a report or analysis of relative strength and sprinting performance.

4.3 Change of Direction

For the purposes of this review, relationships between strength and COD performance were evaluated strictly on pre-planned COD tests because the neuromuscular strategies associated with agility (reactive) performance are unique and highly dependent on a combination of cognitive processing strategies [136]. Thus, this review focused on the relationship between the physical capacity of COD and the physical attribute of strength. However, future research should seek to understand the interaction between perceptual-cognitive strategies and the ability to use physical attributes such as strength during agility tasks as the most recent research indicates that direct relationships between strength and agility are only small in magnitude or do not differ between stronger and weaker athletes [137, 138]. Similar to sprinting, RFD is critical for COD tasks that occur in periods that preclude athletes from producing their maximal force capacity. Specifically, the plant phase, which is when the actual COD occurs, can range from 0.23–0.77 s dependent on the entry velocity and severity of the COD angle required [137, 139–141]. All ground contact lengths during a COD exceed the typical ground contact time of both the acceleration phase of sprinting (0.17–0.2 s) [142] and the maximal velocity phase of sprinting (0.09–0.11 s) [143]. Therefore a strong relationship between maximal strength and COD performance would be expected, as there is greater amount of time available to utilize one's maximal strength. However, similar to sprinting, COD performance requires not only having the strength to change one's momentum, but also the ability to use this strength through coordinated body movements within the constraints of the activity [136, 137, 139, 144, 145].

Based upon mathematical principles, those that can apply greater force over a given time (greater impulse) should be able to accelerate or change momentum with the fastest velocity. However, the disparity in the expected magnitude of the relationship between strength and COD may have more to do with the tests used to measure "COD ability" and "strength" rather than the lack of association between strength and COD ability. This hypothesis is supported by research questioning the validity of "total time" in the assessment of COD ability and that smaller

Table 4 Summary of studies correlating maximal strength and sprinting performance variables

Study	Subjects (n)	Strength measure	Sprint measure	Correlation results
Baker and Nance [126]	Professional male rugby league players ($n = 20$)	3RM BS, 3RM HPC, Rel 3RM BS, Rel 3RM HPC	10, 40 m times	10 m: $r = -0.06$ (3RM BS), $r = -0.36$ (3RM HPC), $r = -0.39$ (Rel 3RM BS), $r = -0.56$ (Rel 3RM PC) 40 m: $r = -0.19$ (3RM BS), $r = -0.24$ (3RM HPC), $r = -0.66$ (Rel 3RM BS), $r = -0.72$ (Rel 3RM PC)
Chaouachi et al. [127]	Tunisian national team male basketball players ($n = 14$)	1RM HS	5, 10, 30 m times	$r = -0.63$ (5 m), $r = -0.68$ (10 m), $r = -0.65$ (30 m)
Comfort et al. [103]	Well-trained youth male soccer players ($n = 34$)	1RM BS, Rel 1RM BS	5, 20 m times	1RM BS: $r = -0.60$ (5 m), $r = -0.65$ (20 m) Rel 1RM BS: $r = -0.52$ (5 m), $r = -0.67$ (20 m)
Cronin and Hansen [81]	Professional male rugby league players ($n = 16$)	3RM BS	5, 10, 30 m times	$r = -0.05$ (5 m), $r = -0.01$ (10 m), $r = -0.29$ (30 m)
Harris et al. [128]	Male national-level rugby training squad and national rugby league premier squad members ($n = 30$)	1RM machine hack squat, Rel 1RM machine hack squat	10, 30/40 m times	10 m: $r = 0.20$ (1RM), $r = -0.10$ (Rel 1RM) 30/40 m: $r = -0.14$ (1RM), $r = -0.33$ (Rel 1RM)
Lockie et al. [129]	Male field sport athletes ($n = 20$)	3RM BS, Rel 3RM BS	0–5, 5–10, and 0–10 m velocity	3RM BS: $r = 0.43$ (0–5 m), $r = 0.60$ (5–10 m), $r = 0.47$ (0–10 m) Rel 3RM BS: $r = 0.50$ (0–5 m), $r = 0.66$ (5–10 m), $r = 0.56$ (0–10 m)
McBride et al. [130]	NCAA division I AA football players ($n = 17$)	Rel 1RM BS	5, 10, 40 m times	$r = -0.45$ (5 m), $r = -0.54$ (10 m), $r = -0.60$ (40 m)
Meckel et al. [131]	NCAA division I female track field sprinters ($n = 30$)	Rel 1RM BS	100 m	$r = -0.89$
Nimphius et al. [42]	Female Australian Institute of Sport state softball players ($n = 10$)	Rel 1RM BS	10 m split to first base, first base, Second base times	10 m split: $r = -0.87$ (Pre-season), $r = -0.85$ (Mid-season), $r = -0.75$ (Post-season) First base: $r = -0.84$ (Pre-season), $r = -0.84$ (Mid-season), $r = -0.80$ (Post-season) Second base: $r = -0.84$ (Pre-season), $r = -0.79$ (Mid-season), $r = -0.83$ (Post-season)
Peterson et al. [83]	First-year male and female collegiate athletes ($n = 55$)	1RM BS, Rel 1RM BS	20, 40 y velocities	1RM BS and 20y: $r = 0.82$ (All), $r = 0.39$ (Males), $r = 0.38$ (Females); 40y: $r = 0.85$ (All), $r = 0.43$ (Males), $r = 0.40$ (Females) Rel 1RM BS and 20y: $r = 0.88$ (All), $r = 0.65$ (Males), $r = 0.72$ (Females); 40y: $r = 0.88$ (All), $r = 0.72$ (Males), $r = 0.71$ (Females)
Requena et al. [84]	Professional male soccer players ($n = 21$)	1RM COHS, knee extensor MVC, plantar flexor MVC	15 m time	$r = -0.47$ (1RM COHS), $r = -0.42$ (Knee Extensor MVC), $r = -0.35$ (Plantar Flexor MVC)
Seitz et al. [132]	Male junior rugby league players ($n = 13$)	1RM BS, Rel 1RM BS, 1RM PC, Rel 1RM PC	20 m time	$r = -0.60$ (1RM BS), $r = -0.57$ (Rel 1RM BS), $r = -0.62$ (1RM PC), $r = -0.64$ (Rel 1RM PC)

Table 4 continued

Study	Subjects (*n*)	Strength measure	Sprint measure	Correlation results
Thomas et al. [133]	Collegiate male soccer and rugby league players ($n = 14$)	IMTP	5, 20 m times	$r = -0.57$ (5 m), $r = -0.69$ (20 m)
Wisløff et al. [50]	International male soccer players ($n = 17$)	1RM HS	10, 30 m times	$r = -0.94$ (10 m), $r = -0.71$ (30 m)
Young et al. [113]	Australian junior national track and field hurdlers, jumpers, and multi-event athletes ($n = 7$)	IS (120° knee angle)	2.5, 10 m at max speed times	$r = -0.72$ (2.5 m), $r = -0.79$ (10 m at max speed)

1RM one repetition maximum, *3RM* three repetition maximum, *BP* bench press, *BS* back squat, *COHS* concentric-only half-squat, *HPC* hang power clean, *HS* half-squat, *IMTP* isometric mid-thigh clean pull, *IS* isometric squat, *MVC* maximal voluntary contraction, *PC* power clean, *Rel* relative, per kilogram of body mass

time intervals [146–148] or direct measures of center of mass velocity [144, 149] provide more valid assessments of COD ability that may ultimately assist in better understanding the underpinning relationship between strength and COD ability. With respect to the measurement of strength, recent research has shown that measures of eccentric, concentric, dynamic, and isometric strength all contribute to COD performance [150]; however, a majority of research simply measures one "type" of strength. When assessing COD performance by the 505 and T-test which require demanding COD (greater than 75°), eccentric strength contributed the most to COD performance [150]. Therefore, our understanding of the association between strength and COD ability are ever-expanding as we examine more specific or valid measures of each underpinning physical quality. Table 5 displays studies that have examined the relationships between maximal strength and COD performance.

Collectively, the studies displayed in Table 5 reported 45 Pearson correlation coefficients between COD performances (examined by a variety of running based tests) and maximal strength (using a variety of multi-joint assessments). Thirty-five of the correlation magnitudes (78 %) indicated a moderate or greater relationship with strength while 27 (60 %) displayed a large or greater relationship with strength. Previous research that has examined the differences in COD time between stronger and weaker subjects has been mixed [56, 137, 138, 144]. Some studies indicated that individuals who are faster during a COD test possess greater strength compared to those who are slower [137, 138]. Other research indicated that there was no difference between stronger and weaker subjects when total time was assessed [56, 144]. The difference in findings may be attributed to the sensitivity of the measure used to assess COD performance. For example, when COD performance was evaluated by total time to complete a COD task and the exit velocity out of a COD task (a measure specifically evaluating the change of direction step), only exit velocity was significantly faster in the stronger subjects [144]. Overall, a majority of the evidence supports a moderate to very large relationship between maximal strength and COD performance, but the limitations or variety of testing methodologies may primarily explain the various magnitudes of the relationships.

5 Influence of Strength on Specific Sport Skills and Performance

While the transfer of strength to the improvement of force-time characteristics is viewed as a positive adaptation from a theoretical standpoint, the transfer of strength to the actual sport skills and performance of athletes is

Table 5 Summary of studies correlating maximal strength and change of direction variables

Study	Subjects (n)	Strength test	COD test	Correlation results
Chaouachi et al. [127]	Tunisian national team male basketball players ($n = 14$)	1RM HS	T-test	$r = 0.18$
Delaney et al. [151]	Professional rugby league players ($n = 31$)	3RM BS, Rel 3RM BS	505-D time, 505-ND time,	3RM BS: $r = -0.28$ (505-D), $r = -0.21$ (505-ND) Rel 3RM BS: $r = -0.52$ (505-D), $r = -0.56$ (505-ND)
Hori et al. [56]	Semiprofessional male Australian rules football players ($n = 29$)	1RM FS, Rel 1RM FS	modified 505 time	1RM FS: $r = -0.37$ Rel 1RM FS: $r = -0.51$
Jones et al. [152]	University students (mixed gender) with various recreational sporting backgrounds ($n = 38$)	Rel 1RM leg press	505 time	$r = -0.45$
Markovic [153]	Male physical education students ($n = 76$)	1RM BS, IS (120° knee angle)	20-yard shuttle run time, slalom run time	1RM BS: $r = -0.31$ (20-yard shuttle run), $r = -0.21$ (Slalom run) IS: $r = 0.03$ (20-yard shuttle run), $r = 0.08$ (Slalom run)
Nimphius et al. [42]	Female West Australian Institute of Sport softball players ($n = 10$)	Rel 1RM BS	505-D time, 505-ND time	505-D: $r = -0.50$ (Pre-season), $r = -0.75$ (Mid-season), $r = -0.60$ (Post-season) 505-ND: $r = -0.75$ (Pre-season), $r = -0.73$ (Mid-season), $r = -0.85$ (Post-season)
Peterson et al. [83]	First-year male and female collegiate athletes ($n = 55$)	1RM BS, Rel 1RM BS	T-test time	1RM BS & T-test: $r = -0.78$ (All), $r = -0.17$ (Males), $r = -0.41$ (Females) Rel 1RM BS & T-test: $r = -0.81$ (All), $r = -0.33$ (Males), $r = -0.63$ (Females)
Spiteri et al. [150]	Female professional basketball players ($n = 12$)	Rel 1RM BS, Rel Con BS, Rel Ecc BS, Rel IMTP	505 time, T-test time	Rel 1RM BS: $r = -0.80$ (505), $r = -0.80$ (T-test), Rel Con BS: $r = -0.79$ (505), $r = -0.79$ (T-test) Rel Ecc BS: $r = -0.89$ (505), $r = -0.88$ (T-test) Rel IMTP: $r = -0.79$ (505), $r = -0.85$ (T-test)
Spiteri et al. [149]	Stronger ($n = 12$) and weaker ($n = 12$) recreational athletes; mixed gender	IS (unilateral)	45° COD task exit velocity, 45° COD task time	Exit Velocity - Stronger: $r = 0.89$ (force application during COD); $r = 0.95$ (impulse during COD); Weaker: $r = 0.52$ (force application during COD); $r = 0.13$ (impulse during COD) (Time) Stronger: $r =$ -0.37 (force application during COD); $r =$ - 0.48 (impulse during COD); Weaker: $r = -0.32$ (force application during COD); $r = 0.17$ (impulse during COD)
Swinton et al. [154]	Scottish Premier League nonprofessional male rugby union players ($n = 30$)	ALLO 1RM BS, ALLO 1RM deadlift	505 time	ALLO 1RM BS: $r = -0.70$ ALLO 1RM Deadlift: $r = -0.72$
Thomas et al. [133]	Collegiate male soccer and rugby league players ($n = 14$)	IMTP	Modified 505 time	$r = -0.57$
Wisløff et al. [50]	International male soccer players ($n = 17$)	1RM HS	10 m shuttle run	$r = -0.68$

Table 5 continued

Study	Subjects (n)	Strength test	COD test	Correlation results
Young et al. [138]	Community-level male Australian rules football players ($n = 24$)	Rel 3RM HS	45° cut COD task time	$r = -0.20$

1RM one repetition maximum, *3RM* three repetition maximum, *505-D* 505 agility test performed with dominant leg, *505-ND* 505 agility test performed with non-dominant leg, *ALLO* allometrically-scaled, *BP* bench press, *BS* back squat, *COD* change of direction, *Con* concentric-only movement, *Ecc* eccentric-only movement, *FS* front squat, *HS* half-squat, *IMTP* isometric mid-thigh clean pull, *IS* isometric squat, *Rel* relative, per kilogram of body mass

paramount. If the strength characteristics of an athlete did not transfer to the performance of the athletes in their sports or events, sport coaches may be less inclined to incorporate resistance training as a method of preparing their athletes to perform. However, previous literature supports the notion that muscular strength is one of the underlying determinants of strength-power performance [5, 25, 96, 97], but is also associated with enhanced endurance performance [155–158]. Further research has examined the relationships between an athlete's strength and their performance in a variety of sports (Table 6).

The examined studies in Table 6 indicate that stronger athletes outperform their weaker counterparts with regard to both strength-power- and endurance-based sports or events. Collectively, 107 correlation magnitudes were reported with 101 (94 %) displaying a relationship with strength that was moderate or greater and 89 (83 %) displaying a large or greater relationship with strength. In support of these findings, several studies have examined sport performance differences between stronger and weaker subjects. These studies indicated that stronger cyclists had a faster 25-m track cycling time compared to weaker cyclists [37], stronger handball players had a greater standing and 3-step running throwing velocity compared to weaker handball players [93], and that stronger sprinters had a faster 100-m time compared to weaker sprinters [131]. The combined evidence of the comparisons between stronger and weaker athletes provides substantial support that stronger athletes within a relatively homogenous level of skill perform better in comparison to weaker athletes.

6 Influence of Strength on Additional Abilities

In addition to influencing an athlete's force-time characteristics, general sport skills, and specific sport skills, muscular strength may also influence several other training and performance characteristics. Some of the training and performance characteristics that may be influenced by muscular strength are the ability to potentiate when using strength-power potentiation complexes, the magnitude of potentiation that an athlete may achieve, and the reduction of injury risk.

6.1 Potentiation

Much research has investigated the acute effects of strength-power potentiation complexes on an individual's explosive performance. While a number of factors may influence one's ability to realize potentiation [167–169], one factor that may be modified through regular strength training is the individual's strength. In fact, previous

Table 6 Summary of studies correlating maximal strength and specific sport skill performance

Study	Subjects (n)	Strength measure	Performance measure	Correlation results
Beckham et al. [30]	Male and female intermediate to advanced weightlifters ($n = 12$)	IMTP, IMTPa	Snatch, C&J, total	IMTP: $r = 0.83$ (Snatch), $r = 0.84$ (C&J), $r = 0.84$ (Total) IMTPa: $r = 0.62$ (Snatch), $r = 0.60$ (C&J), $r = 0.61$ (Total)
Behm et al. [159]	Secondary school and current/former junior level hockey players ($n = 30$)	Dominant leg 1RM leg press dominant leg Rel 1RM leg press	On-ice skating time	$r = -0.30$ (1RM leg press), $r = -0.31$ (Rel 1RM leg press)
Carlock et al. [44]	National-level male and female junior and senior weightlifters ($n = 64$)	1RM BS, Rel 1RM BS	Snatch, C&J, Rel snatch, Rel C&J	1RM BS & Snatch: $r = 0.93$ (Men), $r = 0.79$ (Women), $r = 0.94$ (All) 1RM BS & C&J: $r = 0.95$ (Men), $r = 0.86$ (Women), $r = 0.95$ (All) 1RM BS: $r = 0.94$ (Snatch), $r = 0.95$ (C&J), $r = 0.38$ (Rel Snatch), $r = 0.36$ (Rel C&J) Rel 1RM BS: $r = 0.46$ (Snatch), $r = 0.50$ (C&J), $r = 0.80$ (Rel Snatch), $r = 0.85$ (Rel C&J)
Dumke et al. [160]	Well-trained male runners ($n = 12$)	IS (140° knee angle)	VO_2 at stages 1–6	$r = 0.57$ (Stage 4), No other data reported
Judge et al. 2011 [161]	Elite and NCAA division I male and female track and field throwers ($n = 57$)	1RM BS	Weight throw personal best	$r = 0.64$ (Males), $r = 0.80$ (Females)
Judge and Bellar [162]	Male and female track and field throwers coached by USA Track and Field level III certified coach ($n = 53$)	1RM BS, 1RM PC, 1RM BP	Shot put season best	$r = 0.77$ (1RM BS), $r = 0.87$ (1RM PC), $r = 0.77$ (1RM BP)
Loturco et al. [105]	Male and female Brazilian national team boxers ($n = 15$)	IHS	Punch impact	$r = 0.68$ (Fixed jab), $r = 0.83$ (Fixed cross), $r = 0.69$ (Self-selected jab), $r = 0.73$ (Self-selected cross)
Reyes et al. [163]	NCAA division III baseball players ($n = 19$)	3RM BP	Bat speed	$r = 0.51$
Speranza et al. [86]	Male first grade ($n = 10$), second grade ($n = 12$), and under 20 s ($n = 14$) semi-professional rugby league players	3RM BS, Rel 3RM BS, 3RM BP, Rel 3RM BP	Rugby tackle ability	3RM BS: $r = 0.67$ (All), $r = 0.72$ (First grade), $r = 0.55$ (Second grade), $r = 0.77$ (Under 20 s) Rel 3RM BS: $r = 0.41$ (All), $r = 0.86$ (First grade), $r = 0.60$ (Second grade), $r = 0.38$ (Under 20 s) 3RM BP: $r = 0.58$ (All), $r = 0.72$ (First grade), $r = 0.18$ (Second grade), $r = 0.70$ (Under 20 s) Rel 3RM BP: $r = 0.23$ (All), $r = 0.27$ (First grade), $r = 0.26$ (Second grade), $r = 0.08$ (Under 20 s)

Table 6 continued

Study	Subjects (n)	Strength measure	Performance measure	Correlation results
Speranza et al. [164]	Male semi-professional rugby league players ($n = 16$)	1RM BS	Rugby tackle performance	$r = 0.71$ (Tackling ability), $r = 0.63$ (Dominant tackles)
Stone et al. [165]	Male and female NCAA division I throwers ($n = 11$)	IMTP	Shot put, weight throw	Shot put: $r = 0.67$ (Pre), $r = 0.71$ (Mid), $r = 0.75$ (Post) Weight throw: $r = 0.70$ (Pre), $r = 0.76$ (Mid), $r = 0.79$ (Post)
Stone et al. [37]	National-level male and female cyclists ($n = 20$)	IMTP, Rel IMTP, IMTPa	Low gear (25 m, curve 1, back stretch, curve 2, finish split times), high gear (25 m, curve 1, back stretch, curve 2, finish split times)	IMTP & Low gear splits: $r = -0.49$ (25 m), $r = -0.54$ (Curve 1), $r = -0.52$ (Back stretch), $r = -0.50$ (Curve 2), $r = -0.50$ (Finish) Rel IMTP & Low gear splits: $r = -0.45$ (25 m), $r = -0.50$ (Curve 1), $r = -0.53$ (Back stretch), $r = -0.54$ (Curve 2), $r = -0.53$ (Finish) IMTPa & Low gear splits: $r = -0.45$ (25 m), $r = -0.50$ (Curve 1), $r = -0.51$ (Back stretch), $r = -0.52$ (Curve 2), $r = -0.51$ (Finish) IMTP & High gear splits: $r = -0.50$ (25 m), $r = -0.51$ (Curve 1), $r = -0.54$ (Back stretch), $r = -0.55$ (Curve 2), $r = -0.54$ (Finish) Rel IMTP & High gear splits: $r = -0.58$ (25 m), $r = -0.60$ (Curve 1), $r = -0.61$ (Back stretch), $r = -0.60$ (Curve 2), $r = -0.58$ (Finish) IMTPa & High gear splits: $r = -0.54$ (25 m), $r = -0.56$ (Curve 1), $r = -0.58$ (Back stretch), $r = -0.57$ (Curve 2), $r = -0.55$ (Finish)
Stone et al. [166]	Male and female national and international level weightlifters ($n = 65$)	1RM BS Rel 1RM BS	Snatch, clean, Rel snatch, Rel clean	1RM BS & Snatch : $r = 0.94$ (All), $r = 0.94$ (Men), $r = 0.79$ (Women) 1RM BS & Clean: $r = 0.95$ (All), $r = 0.95$ (Men), $r = 0.86$ (Women) Rel 1RM BS & Rel Snatch: $r = 0.80$ (All), $r = 0.68$ (Men), $r = 0.71$ (Women) Rel 1RM BS & Rel Clean: $r = 0.85$ (All), $r = 0.73$ (Men), $r = 0.81$ (Women)
Stone et al. [166]	Male and female elite-level American weightlifters ($n = 16$)	IMTP, Rel IMTP, IMTPa	Snatch, C&J, Rel snatch, Rel C&J, SnatchA, C&Ja	IMTP: $r = 0.83$ (Snatch), $r = 0.84$ (C&J) Rel IMTP: $r = 0.37$ (Rel Snatch), $r = 0.24$ (Rel C&J) IMTPa: $r = 0.50$ (SnatchA), $r = 0.50$ (C&Ja)

1RM one repetition maximum, *3RM* three repetition maximum, *BP* bench press, *BS* back squat, *C&J* clean and jerk, *C&Ja* allometrically-scaled clean and jerk, *IMTP* isometric mid-thigh clean pull, *IMTPa* allometrically-scaled isometric mid-thigh clean pull, *IS* isometric squat, *IHS* isometric half-squat, *PC* power clean, *Rel* relative, per kilogram of body mass, *SnatchA* allometrically-scaled snatch research indicated that greater magnitudes of potentiation can be achieved following strength training [170]. This may be attributed to the ability of stronger subjects to develop fatigue resistance to high loads as an adaptation to repeated high load training [171–174]. Additional research examined the relationships between the absolute and relative strength characteristics of subjects and the changes in performance following a potentiation protocol (Table 7).

Collectively, the studies displayed in Table 7 reported 67 Pearson correlation coefficients. Of those reported, 39 (58 %) displayed a moderate or greater relationship with strength, while 33 (49 %) displayed a correlation magnitude that was large or greater. In support of these findings, a number of studies have indicated that stronger subjects potentiate earlier [172, 185, 187] and to a greater extent [172, 178, 184–187, 192–195] compared to their weaker counterparts. However, other studies noted no statistical differences in the potentiation displayed between strong and weak subjects [196–198]. A possible explanation for the results of the latter studies may be the design of the examined strength-power potentiation complexes. Two of the studies [196, 197] did not report any statistical increases in vertical jump performance following the examined potentiation protocols, making comparisons between stronger and weaker subjects challenging. The remaining study [198] did not find any statistical differences within the stronger and weaker groups following the implemented potentiation protocols compared to the performances following the control protocol used. While relative strength is a major contributing factor to the timing and magnitude of potentiation, the design of the strength-power potentiation complex cannot be overlooked as it ultimately produces a state of preparedness for subsequent activity [168]. A second explanation for the lack of statistical differences between stronger and weaker subjects may be the range of the subjects' abilities within each group. For example, males and females were grouped together in one study when comparing potentiation differences between stronger and weaker subjects [197], while large standard deviations within groups may have prevented statistical differences from being found in another study [196].

Collectively, the previous literature indicates that by achieving greater strength, an individual may be able to realize potentiation effects at an earlier rest interval and to a greater extent. From a practical standpoint, some authors have noted that those with the ability to back squat at least twice their body mass to either parallel depth [184, 185, 187] or to 90° of knee flexion [199] may have a greater potential to potentiate their performance as compared to their weaker counterparts. Similarly, Berning et al. [193] indicated that a level of strength required to achieve greater magnitudes of potentiation is the ability to back squat at least 1.7 times one's body mass to parallel depth.

6.2 Injury Rate

Previous research has indicated that muscular strength may be as important as anaerobic power for performance as well as injury prevention in soccer players [200]. Along with winning, the rate of injuries in sports and training is one of the primary concerns of athletes, coaches, and practitioners. If athletes are injured in some capacity, they cannot contribute to the overall performance of the team on the field or court. From a coaching perspective, the introduction of new training modalities may not be well received because certain exercises are viewed as injurious. However, an appropriate and progressive prescription, using a variety of methods that focus on improving strength, may decrease the overall occurrence of injuries. Previous research has indicated that there was a decrease in the injury rate per 1,000 exposure hours in collegiate soccer players following the addition of a strength training program [201]. In addition, Sole et al. [202] indicated that the greatest value of team isometric mid-thigh pull strength coincided with the lowest annual injury rate experienced in female volleyball players. This evidence lends support to the idea that increases in strength may play an important role in reducing the occurrence of injuries. Several other studies [200, 203, 204] and reviews [205–208] support this concept. In fact, a meta-analysis indicated that the examined strength training protocols reduced sports injuries to less than one-third and that overuse injuries could be almost halved [207]. Resistance training may reduce the number of injuries due to increases in the structural strength of ligaments, tendons, tendon to bone and ligament to bone junctions, joint cartilage, and connective tissue sheaths within muscles [205]. Moreover, positive changes in bone mineral content as a result of resistance training may aid in the reduction of skeletal injuries. Collectively, the previous literature indicates that resistance training is a modality that may decrease injury rates and that stronger athletes are less likely to get injured. Therefore, a primary focus of strength and conditioning practitioners may be to increase the overall strength of their athletes in order to not only increase performance, but to also decrease the likelihood of an injury occurring.

7 Testing and Monitoring Strength Characteristics

Regular testing and monitoring of an athlete's performance may be the most effective way to provide useful information to the sport or event coaches about the athlete's training state [209, 210]. Moreover, this information can be used to prescribe and adapt training programs to provide an optimal training stimulus for athletes. With regard to

Table 7 Summary of studies correlating maximal strength and potentiation effects

Study	Subjects (n)	Strength measure	Potentiation test	Correlation results
Bellar et al. [175]	NCAA division I male and female track and field throwers ($n = 17$)	1RM PC 1RM BS	Weight throw distance	$r = 0.54$ (1RM PC), $r = 0.22$ (1RM BS)—Overweight 1 $r = 0.55$ (1RM PC), $r = 0.23$ (1RM BS)—Overweight 2
Bevan et al. [176]	Professional rugby players ($n = 26$)	3RM BP	BP throw	$r = 0.52$ (8 min)
Chaouachi et al. [177]	Elite male volleyball players ($n = 12$)	1RM HS	CMJ height, PP, PF, PV, PPave	$r = -0.02$–0.12 (Time to max JH, PP, PF, PV, PPave) $r = -0.07$–0.12 (Potentiation response of JH, PP, PF, PV, PPave)
Duthie et al. [178]	Female hockey and softball players ($n = 11$)	1RM HS	Jump squat PP, PF	$r = 0.66$ (PP), $r = 0.76$ (PF)
Jo et al. [172]	Recreationally-trained men ($n = 12$)	Rel 1RM BS	Wingate time to PPmax	$r = -0.77$
Judge et al. [179]	NCAA division I male and female track and field throwers ($n = 41$)	1RM PC 1RM BS 1RM BP	Overhead back shot put throw	$r = 0.34$ (1RM PC), $r = 0.30$ (1RM BS), $r = 0.27$ (1RM BP)—heavy condition $r < 0.16$ (1RM PC, BS, BP)—light condition $r < 0.16$ (1RM PC, BS, BP)—control condition
Kilduff et al. [180]	Professional rugby players ($n = 23$)	3RM BS Rel 3RM BS	CMJ PP	3RM BS: $r = 0.56$ (8 min), 0.63 (12 min) Rel 3RM BS: $r = 0.63$ (12 min)
Kilduff et al. [180]	Professional rugby players ($n = 23$)	3RM BP Rel 3RM BP	BP throw PP	3RM BP: $r = 0.59$ (12 min) Rel 3RM BP: $r = 0.21$ (12 min)
Kilduff et al. [181]	Professional rugby players ($n = 20$)	3RM BS	CMJ height	$r = 0.49$ (8 min)
Mangus et al. [182]	Male weightlifters ($n = 11$)	Rel HS Rel QS	CMJ height	$r = -0.14$ (Rel HS), $r = -0.17$ (Rel QS)
Okuno et al. [183]	Male handball players ($n = 12$)	1RM HS	RSAbest, RSAave, RSAindex	$r = 0.03$ (RSAbest), $r = 0.50$ (RSAave), $r = 0.56$ (RSAindex)
Ruben et al. [184]	Resistance-trained men ($n = 12$)	1RM BS	Horizontal plyometric hurdle hops	$r = 0.82$ (PPave), $r = 0.53$ (PFave), $r = 0.70$ (PVave), $r = 0.81$ (PPmax), $r = 0.30$ (PFmax), $r = 0.77$ (PVmax)
Seitz et al. [185]	Male junior rugby league players ($n = 18$)	Rel 1RM BS	CMJ PP, time to PPmax, JH, time to JHmax	$r = 0.78$ (PP), $r = -0.69$ (Time to PPmax) $r = 0.74$ (JH), $r = -0.69$ (Time to JHmax)
Seitz et al. [132]	Male junior rugby league players ($n = 13$)	Rel 1RM BS Rel 1RM PC	20 m sprint potentiation response	$r = 0.56$ (Rel 1RM BS), $r = 0.63$ (Rel 1RM PC)
Suchomel et al. [186]	Resistance-trained men ($n = 15$)	Rel 1RM BS Rel 1RM COHS	SJ height potentiation response	Rel 1RM BS: $r = 0.52$ (Ballistic), $r = 0.63$ (Non-ballistic) Rel 1RM COHS: $r = 0.57$ (Ballistic), $r = 0.48$ (Non-ballistic)
Suchomel et al. [187]	Resistance-trained men ($n = 16$)	Rel 1RM BS Rel 1RM COHS	SJ height maximum potentiation response	Rel 1RM BS: $r = 0.64$ (Ballistic), $r = 0.54$ (Non-ballistic) Rel 1RM COHS: $r = 0.74$ (Ballistic), $r = 0.47$ (Non-ballistic)
Terzis et al. [188]	Male and female physical education students ($n = 16$)	6RM leg press	Underhand shot throw	$r = 0.50$

Table 7 continued

Study	Subjects (*n*)	Strength measure	Potentiation test	Correlation results
Tsolakis et al. [189]	Male and female international level fencers ($n = 23$)	1RM leg press	CMJ PP	$r = -0.55$ (12 min)
West et al. [190]	Professional rugby players ($n = 20$)	1RM BP	BP throw PP	$r = 0.63$ (Ballistic), $r = 0.68$ (Heavy resistance training)
Witmer et al. [191]	Male and female collegiate athletes and recreationally-trained athletes ($n = 24$)	1RM BS	CMJ height, vertical stiffness	CMJ height: $r = -0.54$ (Men), 0.10 (Women) CMJ vertical stiffness: $r = -0.43$ (Men), -0.36 (Women)
Young et al. [192]	Recreationally-trained men ($n = 10$)	5RM HS	Loaded CMJ height	$r = 0.73$

RM repetition maximum, *BP* bench press, *BS* back squat, *CMJ* countermovement jump, *COHS* concentric-only half-squat, *HS* half-squat, *IS* isometric squat, *PC* power clean, *PF* peak force, *PP* peak power, *PV* peak velocity, *QS* quarter-squat, *Rel* relative, per kilogram of body mass, *RSA* repeated sprint ability, *SJ* squat jump

testing and monitoring an athlete's strength, sport scientists and practitioners may use various tests to examine an athlete's isometric, dynamic, and reactive strength characteristics. The subsequent paragraphs briefly discuss previous research that has used isometric, dynamic, and reactive strength testing to examine the strength characteristics of individuals. For a thorough review on different methodologies of strength assessment, readers are directed to McMaster and colleagues [211].

Regular monitoring can also assist in better understanding the aforementioned relationships between maximal strength and performances, as the required motor learning strategies to manifest improvements in overall strength into skilled performance must be recognized. The delay between increased physical capacity and ability to actualize increased strength into improved performance is termed lag time [48, 212]. The concept of lag time, or the length of time it takes for an athlete to "learn to utilize their new found strength," is important to consider when trying to determine the transfer of training effect from one underlying physical attribute to an athletic skill such as sprinting and jumping. Thus, regular testing and assessment of the data is critical in order to assess or determine the lag within various activities.

7.1 Isometric Strength

As displayed in the tables above, many studies have assessed the maximum strength of subjects by using an isometric strength test such as the isometric mid-thigh pull, isometric squat, or isometric half-squat. While these tests do not provide a maximum load lifted, previous research has displayed notable relationships between the isometric strength tests and dynamic strength performance [29, 36, 106]. In addition to examining relationships between maximal isometric strength and various performance characteristics (Tables 1, 2, 3, 4, 5, 6, 7), isometric strength tests have been used to examine different phases of an exercise [213], the effect of a training program on muscular strength characteristics [29, 214], and determine force production differences among athletic teams [215]. The versatility of an isometric strength test should not be overlooked. Isometric strength tests are time efficient, particularly with large groups, and may provide a truer measure of "maximum" strength compared to dynamic strength testing in which the final load attempted may be overestimated. However, as with any maximal strength test, isometric strength tests should be used sparingly as they can be taxing for the individual and may require the need to slightly modify training during the day of testing.

Sport scientists and practitioners must keep in mind the sport specificity of the athlete when using isometric testing. In other words, the athlete must be tested in a position that is related to the success of their sport. For example, previous research has indicated that the greatest amount of force and power is produced during the second pull of weightlifting movements [216]. Thus, it would be logical to test weightlifters in a position that is specific to the second pull, as demonstrated by previous research [30, 166, 217]. Another example may be testing sprinters or bobsledders at hip and knee angles that correspond to different phases of speed development (i.e., acceleration, transition, velocity, competition speed) [218]. By testing the individual during each phase, the sport scientist and coach will receive information about the strengths and weaknesses of an athlete's overall sprinting performance. From here, modifications to the individual's training program can be made to eliminate any potential weaknesses. However, sport scientists and practitioners should keep in mind that performing such tests should not hinder athletes from completing their planned training program.

7.2 Dynamic Strength

While isometric strength testing has its advantages, so too does dynamic strength testing. Dynamic strength testing may be the most common method of measuring an individual's strength. This is typically accomplished by having the individual perform a repetition maximum (RM) test, where the individual lifts as much weight as possible for a specific number of repetitions. Examples listed in the Tables 1, 2, 3, 4, 5, 6, 7 include RM tests ranging from 1RM–6RM tests of either the back squat, front squat, half-squat, power clean, hang clean, leg press, or bench press. While the previous exercises have both eccentric and concentric muscle actions, additional studies have used concentric-only movements [186, 187, 219] or eccentric-only movements [150] to assess maximal strength characteristics within each of the muscle actions individually that comprise overall dynamic strength.

Dynamic strength tests may be viewed as more relevant to an athlete's abilities due to their similarities to movements completed in various sports or events. Previous research has used dynamic strength tests to examine the effect of specific training programs [124, 220], the effect that a competitive season had on muscular strength [221–224], and contributing factors that affect COD performance [150]. Similar to isometric strength testing, dynamic strength testing should be completed sparingly due to its taxing nature. While some practitioners use dynamic strength 1RM tests to prescribe training loads, others may discourage the practice of "maxing out." An alternative option for the latter practitioners would be to estimate an individual's 1RM using the set-rep best method described by Stone and O'Bryant [225]. The set-rep best method uses loads performed in training for a specific repetition scheme and estimates training loads for other repetitions schemes, but also a 1RM. This approach may be applied to any exercise, but may be the most useful for exercises that do not have specific criteria for a successful 1RM attempt, such as weightlifting pulling derivatives [226–236].

7.3 Reactive Strength

Reactive strength can be described as the ability of an athlete to change quickly from an eccentric to concentric muscular contraction [237]. The two primary methods of assessing reactive strength are through performing either drop jumps or countermovement jumps to calculate the variables reactive strength index (RSI; drop jump height $\times$ ground contact time^{-1}) or reactive strength index-modified (RSImod; countermovement jump height $\times$ time to takeoff^{-1}), respectively. Although different from maximal isometric and dynamic strength testing, previous research has indicated that there are strong relationships between maximal isometric strength and RSImod [238]. In addition, reactive strength testing can provide further information to the practitioners regarding how an individual achieves a certain standard of dynamic performance. For example, previous research examining RSI has determined that it is a reliable performance variable [239], can differentiate between field athletes with higher or lower acceleration abilities [129], can be used to monitor neuromuscular fatigue [240], and can be used as an indicator of the current training conditions [241]. Additional research has determined that RSImod is a reliable performance variable that can be used to monitor explosive performance acutely [242, 243], but also over the course of a competitive season [244]. Furthermore, RSImod can distinguish performance differences between teams [245], within teams [246], and can be used to assess an athlete's ability to effectively use the stretch-shortening cycle to achieve a specific jump height [247]. While scientific equipment is needed to assess RSI and RSImod [247], more information can be gathered that will provide practitioners with greater understanding of an individual's current performance capacity.

8 Absolute and Relative Standards of Strength

While absolute strength may be the deciding factor of which athlete is victorious in some sports (e.g. linemen in American football), the relative strength of an individual may be more important in certain sports where one must move their own body mass (e.g., track and field sprinting and jumping) or is competing in a sport that has weight class divisions (e.g., weightlifting). At present, no scales exist that recommend certain standards of relative strength

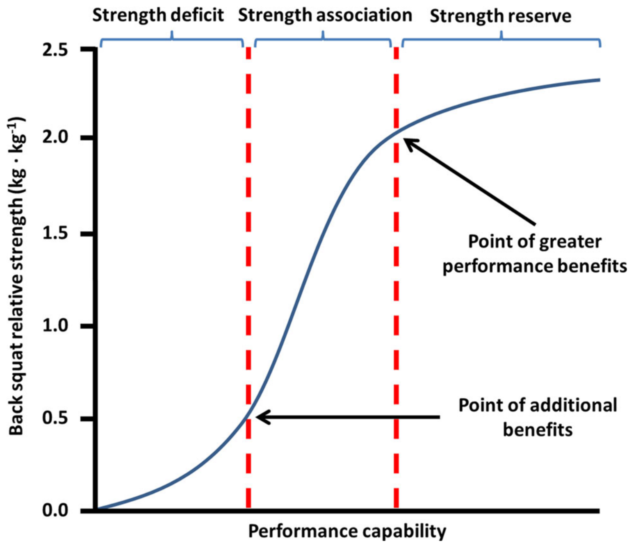

Fig. 1 Theoretical relationship between back squat relative strength and performance capability

for individuals in different sports; however, general recommendations can be made based on existing literature. Previous research has suggested that individuals who back squatted at least twice their body mass produced greater external mechanical power during a vertical jump [5, 12], sprinted faster and jumped higher [50], and potentiated earlier [185, 187] and to a greater extent [184, 185, 187] compared to individuals who did not. Figure 1 illustrates the theoretical relationship between relative back squat strength (per kilogram of body mass) and performance capabilities. It should be noted that this model is specific to the back squat based on the findings of the research presented earlier in this paragraph and the regular use of the back squat as a standard measure of strength. Moreover, the theoretical nature of the presented model should be emphasized. While a number of studies indicate that the ability to back squat at least twice one's body mass is indicative of a greater performance, information regarding specific standards of required strength is still lacking. The presented model indicates that there are three primary strength phases including strength deficit, strength association, and strength reserve. Previous work by Keiner et al. [248] provides a timeline for the presented model by suggesting that with 4–5 years of structured strength training, relative strength levels with the back squat should be at a minimum 2.0 for late adolescents (16–19 years old), 1.5 for adolescents (13–15 years old), and 0.7 for children (11–12 years old) (Fig. 2).

8.1 Strength Deficit Phase

The strength deficit phase may be the shortest phase based on the motor learning capacity of the individual. This phase suggests that although an individual is improving their

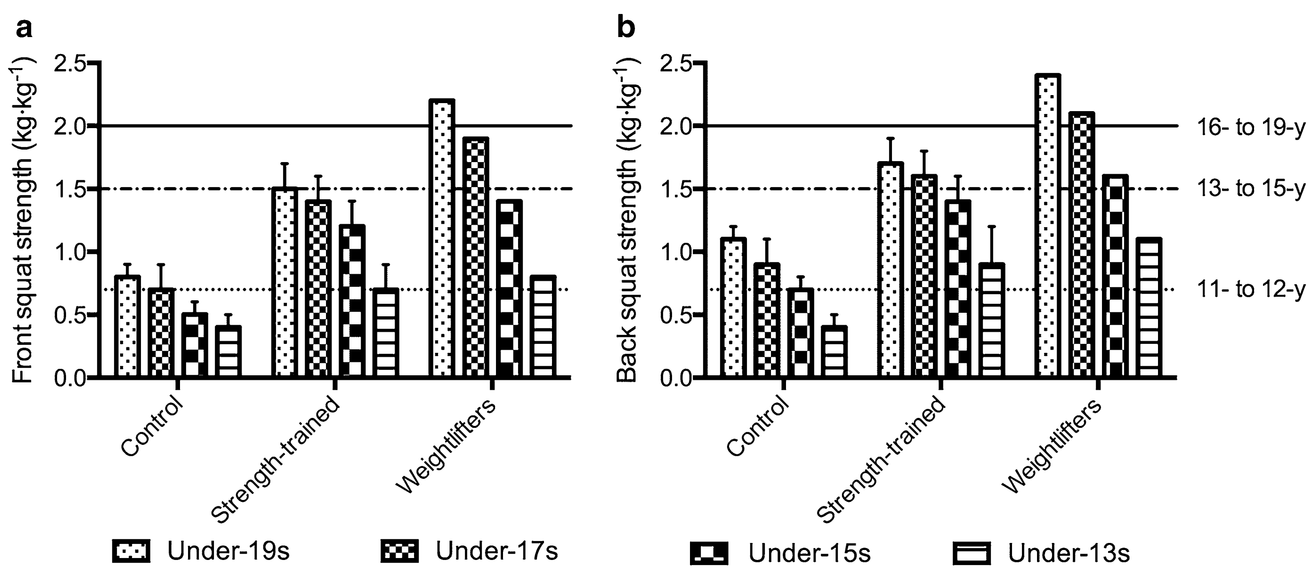

Fig. 2 Relative front squat (**a**) and back squat strength (**b**) comparison between control subjects (2 years of soccer training only; mean and standard deviation [SD]), strength-trained subjects (2 years soccer and strength training; mean and SD), and young elite weightlifters (mean only). Values for the weightlifters represent predicted one-repetition maximum (1RM) of the front squat and back squat based on their 5RM strength testing of the weight classes closest to that of the soccer players in each age group [281]. *Notes*: Weightlifters were tested with full depth squats and all soccer players (control- and strength-trained) were tested with parallel depth squats. *Lines* are drawn at the recommended standards of strength for young elite athletes with long-term training (a training age commensurate with appropriate resistance training from 7 or 8 years of age). Figure created by the authors from data in Keiner et al. [248]

strength (i.e., their ability to generate force), they may not be able to exploit their levels of strength and translate them into positive performance benefits in their respective sport. This is supported by the phasic progression concepts from previous literature [25, 63, 64] that indicates that central and local factors (i.e., motor unit recruitment, fiber type, and co-contraction) enhance the ability to increase maximum strength. Novice athletes within this phase are often going through stages of physical literacy, especially if they have not been previously exposed to strength training [249, 250]. The strength deficit phase will ultimately continue until the individual becomes competent with the strength training exercise.

8.2 Strength Association Phase

As the athlete gets stronger, he or she enters the strength association phase where increases in strength often directly translate to an improved performance. As indicated in the model, this phase is characterized by a nearly linear relationship between relative strength and performance capability. Specifically, further increases in maximum strength combined with central factors, the specificity of the task, and the coordination of multiple joints enhance an individual's ability to increase muscular performance [25, 63, 64]. The duration of this phase may be based primarily on two physiological mechanisms including muscle cross-sectional area or architectural changes and supraspinal/spinal neuromuscular adaptations that occur as a result of regular strength training. Specifically, the cross-sectional area or architectural changes that are characteristic of strength training are greater Type II/I functional cross-sectional area [251–253] and pennation angle changes [254–256]. The supraspinal/spinal neuromuscular adaptations include increases in motor unit rate coding [28, 257], neural drive [71, 258–260], inter- and possibly intra-muscular coordination [261–267], motor unit synchronization [268, 269], and the ability to use the stretch-shortening cycle, while decreasing neural inhibitory processes [63, 270]. Previous studies that have examined training for maximal strength have reported changes in muscle architecture after 4–5 weeks [271, 272] and increased tendon stiffness after 9–10 weeks [273, 274]. As changes in muscle architecture [20, 254] and tendon stiffness [275, 276] may affect the electromechanical delay and rate of force development during stretch-shortening cycle tasks, it is important to note the time needed for positive training adaptations to occur.

8.3 Strength Reserve Phase

The final phase of the proposed model is the strength reserve phase. Athletes who reach this phase have dramatically improved their ability to produce force primarily due to local and central adaptations and alterations in task specificity [25, 277, 278]. During the strength reserve phase, athletes may continue to gain relative strength; however, the direct benefits to performance may not be as substantial. In fact, a previous review indicated that while strength is a basic quality that influences an athlete's performance, the degree of this influence may diminish when athletes maintain a very high level of strength [279]. Thus, the window of adaptation for further strength enhancement is reduced as an individual increases their maximal strength. This may be why other literature has suggested that the emphasis of training may be shifted towards "power" or RFD training after a specific standard of strength has been achieved [25, 68, 279, 280]. That is not to say that individuals should not seek to continue improving their strength, rather stronger individuals can focus more on maintaining their strength, while placing more emphasis on RFD and speed adaptations. It should be noted, however, that limited research has examined the differences in performance between individuals that can squat greater than or equal to 2.5× their body mass versus 2.0× and 1.5×. Moreover, no research has discussed the changes in performance after transitioning from a 2.0× to a 2.5× body mass squat.

9 Limitations

The current review was primarily descriptive to provide a comprehensive description with as much of the literature represented as possible. The benefit of such a comprehensive description results in the limitation that a full meta-analytical review could come to stronger conclusions. However, each area of the current review would require a separate meta-analysis and therefore would suffer from not being able to draw on the multi-factorial discussion presented in the current review. Furthermore, it should be noted that much of the interpretation of existing studies came from correlational analyses and the readers should consider that correlation does not necessarily indicate causation.

10 Conclusions

While certain underlying factors of an athlete's performance cannot be manipulated (e.g., genetics), sport scientists and practitioners can manipulate an athlete's absolute and relative strength with regular strength training. Greater muscular strength can enhance the force-time characteristics (e.g., RFD and external mechanical power) of an individual that can then translate to their athletic performance. Muscular strength is strongly correlated to superior jumping, sprinting, COD, and sport-specific performance. Additional benefits of stronger individuals include the ability to take advantage of postactivation potentiation and a decreased injury rate. Sport scientists and practitioners may monitor the isometric, dynamic, and reactive strength of individuals in order to provide optimal training stimuli to enhance specific strength characteristics that translate to performance. It is recommended that athletes should strive to become as strong as possible within the context of their sport or event. Regarding relative lower body strength, it appears that the ability to back squat at least twice one's body mass may lead to greater athletic performance compared to those who possess lower relative strength. The vast majority of the literature supports the notion that stronger athletes demonstrate superior RFD and external mechanical power, and subsequently jump higher, run faster, perform COD tasks faster, potentiate earlier and to a greater extent, and are less likely to get injured. Therefore, sport scientists and practitioners could conclude that there may be no substitute for greater muscular strength as it underpins a vast number of attributes that are related to improving an individual's performance across a wide range of both general and sport specific skills while simultaneously reducing their risk of injury when performing these skills.

Despite the information described in this review, a number of research questions regarding the influence of strength on an athlete's overall performance still exist. Information regarding specific standards of required strength is still lacking. While a number of studies indicate that the ability to back squat at least twice one's body mass is indicative of a greater performance, no research has established standards for greater performance using isometric strength measurements. Furthermore, no levels of relative upper body strength that display a superior performance compared to lower relative strength have been reported. While general conclusions can be made with regard to the influence of strength on an athlete's performance, more research is needed with female athletes with regard to how their relative strength levels relate to their performance. Additional research with female athletes would allow for more specific recommendations to be made. The studies discussed within this review focused on bilateral strength measures, primarily because it may not be practical to perform a 1RM test with a single limb. However, due to the unilateral nature of certain sports and events (e.g., sprint events, hockey, etc.), further research examining the transfer of bilateral strength to single leg force-time characteristics and the transfer of single-leg strength training to bilateral force-time characteristics, strength, and overall performance is needed. Finally, future research should examine the effect of longitudinal resistance training, particularly with respect to long-term athlete development over several years to gain a better understanding of the influence of strength on the development and performance of an athlete.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of interest Timothy Suchomel, Sophia Nimphius, and Michael Stone declare that they have no conflicts of interest relevant to the content of this review.

References

1. Stone MH. Position statement: explosive exercises and training. Natl Strength Cond Assoc J. 1993;15(3):7–15.
2. Siff M. Biomechanical foundations of strength and power training. In: Zatsiorsky V, editor. Biomechanics in Sport. London: Blackwell Scientific Ltd; 2001. p. 103–39.
3. Hopkins WG. A scale of magnitude for effect statistics. http://sportsci.org/resource/stats/effectmag.html. 2014.
4. Baker D. A series of studies on the training of high-intensity muscle power in rugby league football players. J Strength Cond Res. 2001;15(2):198–209.
5. Stone MH, Moir G, Glaister M, et al. How much strength is necessary? Phys Ther Sport. 2002;3(2):88–96.
6. Morrissey MC, Harman EA, Johnson MJ. Resistance training modes: specificity and effectiveness. Med Sci Sports Exerc. 1995;27(5):648–60.
7. Young WB, Newton RU, Doyle TLA, et al. Physiological and anthropometric characteristics of starters and non-starters and playing positions in elite Australian Rules football: a case study. J Sci Med Sport. 2005;8(3):333–45.
8. Iguchi J, Yamada Y, Ando S, et al. Physical and performance characteristics of Japanese division 1 collegiate football players. J Strength Cond Res. 2011;25(12):3368–77.
9. Gabbett T, Kelly J, Ralph S, et al. Physiological and anthropometric characteristics of junior elite and sub-elite rugby league players, with special reference to starters and non-starters. J Sci Med Sport. 2009;12(1):215–22.
10. Fry AC, Kraemer WJ. Physical performance characteristics of American collegiate football players. J Strength Cond Res. 1991;5(3):126–38.
11. Gabbett TJ. Physiological and anthropometric characteristics of starters and non-starters in junior rugby league players, aged 13–17 years. J Sports Med Phys Fit. 2009;49(3):233–9.
12. Barker M, Wyatt TJ, Johnson RL, et al. Performance factors, psychological assessment, physical characteristics, and football playing ability. J Strength Cond Res. 1993;7(4):224–33.
13. Sands WA, Smith LS, Kivi DM, et al. Anthropometric and physical abilities profiles: US National Skeleton Team. Sports Biomech. 2005;4(2):197–214.
14. Baker D, Newton RU. Comparison of lower body strength, power, acceleration, speed, agility, and sprint momentum to describe and compare playing rank among professional rugby league players. J Strength Cond Res. 2008;22(1):153–8.
15. Baker D. Comparison of upper-body strength and power between professional and college-aged rugby league players. J Strength Cond Res. 2001;15(1):30–5.

16. Clark KP, Weyand PG. Are running speeds maximized with simple-spring stance mechanics? J Appl Physiol. 2014;117(6):604–15.
17. le Gall F, Carling C, Williams M, et al. Anthropometric and fitness characteristics of international, professional and amateur male graduate soccer players from an elite youth academy. J Sci Med Sport. 2010;13(1):90–5.
18. Hansen KT, Cronin JB, Pickering SL, et al. Do force–time and power–time measures in a loaded jump squat differentiate between speed performance and playing level in elite and elite junior rugby union players? J Strength Cond Res. 2011;25(9):2382–91.
19. Aagaard P, Simonsen EB, Andersen JL, et al. Increased rate of force development and neural drive of human skeletal muscle following resistance training. J Appl Physiol. 2002;93(4):1318–26.
20. Andersen LL, Aagaard P. Influence of maximal muscle strength and intrinsic muscle contractile properties on contractile rate of force development. Eur J Appl Physiol. 2006;96(1):46–52.
21. Garhammer J, Gregor R. Propulsion forces as a function of intensity for weightlifting and vertical jumping. J Strength Cond Res. 1992;6(3):129–34.
22. Thorstensson A, Karlsson J, Viitasalo JHT, et al. Effect of strength training on EMG of human skeletal muscle. Acta Physiol Scand. 1976;98(2):232–6.
23. Zatsiorsky V. Science and practice of strength training. Champaign: Human Kinetics; 1995.
24. Aagaard P. Training-induced changes in neural function. Exerc Sport Sci Rev. 2003;31(2):61–7.
25. Stone MH, Stone M, Sands WA. Principles and practice of resistance training. Champaign: Human Kinetics; 2007.
26. Andersen LL, Andersen JL, Zebis MK, et al. Early and late rate of force development: differential adaptive responses to resistance training? Scand J Med Sci Sports. 2010;20(1):e162–9.
27. Häkkinen K, Komi PV, Alen M. Effect of explosive type strength training on isometric force-and relaxation-time, electromyographic and muscle fibre characteristics of leg extensor muscles. Acta Physiol Scand. 1985;125(4):587–600.
28. van Cutsem M, Duchateau J, Hainaut K. Changes in single motor unit behaviour contribute to the increase in contraction speed after dynamic training in humans. J Physiol. 1998;513(1): 295–305.
29. Bazyler CD, Beckham GK, Sato K. The use of the isometric squat as a measure of strength and explosiveness. J Strength Cond Res. 2015;29(5):1386–92.
30. Beckham GK, Mizuguchi S, Carter C, et al. Relationships of isometric mid-thigh pull variables to weightlifting performance. J Sports Med Phys Fitness. 2013;53(5):573–81.
31. Haff GG, Carlock JM, Hartman MJ, et al. Force-time curve characteristics of dynamic and isometric muscle actions of elite women olympic weightlifters. J Strength Cond Res. 2005;19(4):741–8.
32. Kawamori N, Crum AJ, Blumert PA, et al. Influence of different relative intensities on power output during the hang power clean: identification of the optimal load. J Strength Cond Res. 2005;19(3):698–708.
33. Kawamori N, Rossi SJ, Justice BD, et al. Peak force and rate of force development during isometric and dynamic mid-thigh clean pulls performed at various intensities. J Strength Cond Res. 2006;20(3):483–91.
34. Kraska JM, Ramsey MW, Haff GG, et al. Relationship between strength characteristics and unweighted and weighted vertical jump height. Int J Sports Physiol Perform. 2009;4(4): 461–73.
35. McGuigan MR, Winchester JB, Erickson T. The importance of isometric maximum strength in college wrestlers. J Sports Sci Med. 2006;5(CSSI):108–13.
36. McGuigan MR, Winchester JB. The relationship between isometric and dynamic strength in college football players. J Sports Sci Med. 2008;7(1):101–5.
37. Stone MH, Sands WA, Carlock J, et al. The importance of isometric maximum strength and peak rate-of-force development in sprint cycling. J Strength Cond Res. 2004;18(4):878–84.
38. Thomas C, Jones PA, Rothwell J, et al. An investigation into the relationship between maximum isometric strength and vertical jump performance. J Strength Cond Res. 2015;29(8):2176–85.
39. Zaras ND, Stasinaki AN, Methenitis SK, et al. Rate of force development, muscle architecture, and performance in young competitive track and field throwers. J Strength Cond Res. 2016;30(1):81–92.
40. Bevan HR, Bunce PJ, Owen NJ, et al. Optimal loading for the development of peak power output in professional rugby players. J Strength Cond Res. 2010;24(1):43–7.
41. Hawley JA, Williams MM, Vickovic MM, et al. Muscle power predicts freestyle swimming performance. Br J Sports Med. 1992;26(3):151–5.
42. Nimphius S, McGuigan MR, Newton RU. Relationship between strength, power, speed, and change of direction performance of female softball players. J Strength Cond Res. 2010;24(4):885–95.
43. Sharp RL, Troup JP, Costill DL. Relationship between power and sprint freestyle swimming. Med Sci Sports Exerc. 1981;14(1):53–6.
44. Carlock JM, Smith SL, Hartman MJ, et al. The relationship between vertical jump power estimates and weightlifting ability: a field-test approach. J Strength Cond Res. 2004;18(3):534–9.
45. Blackburn JR, Morrissey MC. The relationship between open and closed kinetic chain strength of the lower limb and jumping performance. J Orthop Sports Phys Ther. 1998;27(6):430–5.
46. Nuzzo JL, McBride JM, Cormie P, et al. Relationship between countermovement jump performance and multijoint isometric and dynamic tests of strength. J Strength Cond Res. 2008; 22(3):699–707.
47. Miyaguchi K, Demura S. Relationships between stretch-shortening cycle performance and maximum muscle strength. J Strength Cond Res. 2008;22(1):19–24.
48. Stone MH, O'Bryant HS, McCoy L, et al. Power and maximum strength relationships during performance of dynamic and static weighted jumps. J Strength Cond Res. 2003;17(1):140–7.
49. Baker D, Nance S. The relation between strength and power in professional rugby league players. J Strength Cond Res. 1999;13(3):224–9.
50. Wisløff U, Castagna C, Helgerud J, et al. Strong correlation of maximal squat strength with sprint performance and vertical jump height in elite soccer players. Br J Sports Med. 2004;38(3):285–8.
51. Moss BM, Refsnes PE, Abildgaard A, et al. Effects of maximal effort strength training with different loads on dynamic strength, cross-sectional area, load-power and load-velocity relationships. Eur J Appl Physiol Occup Physiol. 1997;75(3):193–9.
52. Moir GL, Gollie JM, Davis SE, et al. The effects of load on system and lower-body joint kinetics during jump squats. Sports Biomech. 2012;11(4):492–506.
53. Weyand PG, Sandell RF, Prime DNL, et al. The biological limits to running speed are imposed from the ground up. J Appl Physiol. 2010;108(4):950–61.
54. Weyand PG, Sternlight DB, Bellizzi MJ, et al. Faster top running speeds are achieved with greater ground forces not more rapid leg movements. J Appl Physiol. 2000;89(5):1991–9.
55. Cormie P, McGuigan MR, Newton RU. Influence of strength on magnitude and mechanisms of adaptation to power training. Med Sci Sports Exerc. 2010;42(8):1566–81.
56. Hori N, Newton RU, Andrews WA, et al. Does performance of hang power clean differentiate performance of jumping, sprinting, and changing of direction? J Strength Cond Res. 2008;22(2):412–8.
57. McLellan CP, Lovell DI, Gass GC. The role of rate of force development on vertical jump performance. J Strength Cond Res. 2011;25(2):379–85.

58. Newton RU, Kraemer WJ, Hakkinen K. Effects of ballistic training on preseason preparation of elite volleyball players. Med Sci Sports Exerc. 1999;31:323–30.
59. Salaj S, Markovic G. Specificity of jumping, sprinting, and quick change-of-direction motor abilities. J Strength Cond Res. 2011;25(5):1249–55.
60. Spiteri T, Nimphius S, Wilkie J. Comparison of running times during reactive offensive and defensive agility protocols. J Aust Strength Cond. 2012;20:73–8.
61. McEvoy KP, Newton RU. Baseball throwing speed and base running speed: the effects of ballistic resistance training. J Strength Cond Res. 1998;12(4):216–21.
62. Marques M, Saavedra F, Abrantes C, et al. Associations between rate of force development metrics and throwing velocity in elite team handball players: a short research report. J Hum Kinet. 2011;29:53–7.
63. Minetti AE. On the mechanical power of joint extensions as affected by the change in muscle force (or cross-sectional area), ceteris paribus. Eur J Appl Physiol. 2002;86(4):363–9.
64. Zamparo P, Minetti A, di Prampero P. Interplay among the changes of muscle strength, cross-sectional area and maximal explosive power: theory and facts. Eur J Appl Physiol. 2002;88(3):193–202.
65. Stone MH, O'Bryant H, Garhammer J, et al. A theoretical model of strength training. Strength Cond J. 1982;4(4):36–9.
66. Stone MH, O'Bryant H, Garhammer J. A hypothetical model for strength training. J Sports Med Phys Fitness. 1981;21(4):342–51.
67. DeWeese BH, Hornsby G, Stone M, et al. The training process: Planning for strength–power training in track and field. Part 1: theoretical aspects. J Sport Health Sci. 2015;4(4):308–17.
68. DeWeese BH, Hornsby G, Stone M, et al. The training process: Planning for strength–power training in track and field. Part 2: practical and applied aspects. J Sport Health Sci. 2015;4(4):318–24.
69. Behm DG, Sale DG. Intended rather than actual movement velocity determines velocity-specific training response. J Appl Physiol. 1993;74(1):359–68.
70. Kaneko M, Fuchimoto T, Toji H, et al. Training effect of different loads on the force-velocity relationship and mechanical power output in human muscle. Scand J Sports Sci. 1983;5(2):50–5.
71. McBride JM, Triplett-McBride T, Davie A, et al. The effect of heavy- vs. light-load jump squats on the development of strength, power, and speed. J Strength Cond Res. 2002;16(1):75–82.
72. Stone MH, Johnson RL, Carter DR. A short term comparison of two different methods of resistance training on leg strength and power. Athl Train. 1979;14:158–61.
73. Stowers T, McMillan J, Scala D, et al. The short-term effects of three different strength-power training methods. Strength Cond J. 1983;5(3):24–7.
74. Wilson GJ, Newton RU, Murphy AJ, et al. The optimal training load for the development of dynamic athletic performance. Med Sci Sports Exerc. 1993;25(11):1279–86.
75. Toji H, Kaneko M. Effect of multiple-load training on the force-velocity relationship. J Strength Cond Res. 2004;18(4):792–5.
76. Toji H, Suei K, Kaneko M. Effects of combined training loads on relations among force, velocity, and power development. Can J Appl Physiol. 1997;22(4):328–36.
77. Cormie P, McGuigan MR, Newton RU. Adaptations in athletic performance after ballistic power versus strength training. Med Sci Sports Exerc. 2010;42(8):1582–98.
78. Harris GR, Stone MH, O'Bryant HS, et al. Short-term performance effects of high power, high force, or combined weight-training methods. J Strength Cond Res. 2000;14(1):14–20.
79. Speranza M, Gabbett TJ, Johnston R, et al. The effect of strength and power training on tackling ability in semi-professional rugby league players. J Strength Cond Res. 2015. doi:10.1519/JSC.0000000000001058.
80. Baker D, Nance S, Moore M. The load that maximizes the average mechanical power output during jump squats in power-trained athletes. J Strength Cond Res. 2001;15(1):92–7.
81. Cronin JB, Hansen KT. Strength and power predictors of sports speed. J Strength Cond Res. 2005;19(2):349–57.
82. Jones MT, Jagim AR, Oliver JM. Associations between testosterone, body composition, and performance measures of strength and power in recreational, resistance-trained men. J Athl Enhancement. 2015;4(2).
83. Peterson MD, Alvar BA, Rhea MR. The contribution of maximal force production to explosive movement among young collegiate athletes. J Strength Cond Res. 2006;20(4):867–73.
84. Requena B, González-Badillo JJ, de Villareal ESS, et al. Functional performance, maximal strength, and power characteristics in isometric and dynamic actions of lower extremities in soccer players. J Strength Cond Res. 2009;23(5):1391–401.
85. Sheppard JM, Cronin JB, Gabbett TJ, et al. Relative importance of strength, power, and anthropometric measures to jump performance of elite volleyball players. J Strength Cond Res. 2008;22(3):758–65.
86. Speranza M, Gabbett TJ, Johnston R, et al. Muscular strength and power correlates of tackling ability in semi-professional rugby league players. J Strength Cond Res. 2015;29(8):2071–8.
87. Bourque PJ. Determinants of load at peak power during maximal effort squat jumps in endurance-and power-trained athletes [Dissertation]. Fredericton (NB): University of New Brunswick (Canada); 2003.
88. Baker D, Newton RU. Adaptations in upper-body maximal strength and power output resulting from long-term resistance training in experienced strength-power athletes. J Strength Cond Res. 2006;20(3):541–6.
89. Cormie P, McBride JM, McCaulley GO. Power-time, force-time, and velocity-time curve analysis of the countermovement jump: impact of training. J Strength Cond Res. 2009;23(1):177–86.
90. McBride JM, Triplett-McBride T, Davie A, et al. A comparison of strength and power characteristics between power lifters, Olympic lifters, and sprinters. J Strength Cond Res. 1999;13(1):58–66.
91. Stoessel L, Stone MH, Keith R, et al. Selected physiological, psychological and performance characteristics of national-caliber United States women weightlifters. J Strength Cond Res. 1991;5(2):87–95.
92. Nuzzo JL, McBride JM, Dayne AM, et al. Testing of the maximal dynamic output hypothesis in trained and untrained subjects. J Strength Cond Res. 2010;24(5):1269–76.
93. Gorostiaga EM, Granados C, Ibanez J, et al. Differences in physical fitness and throwing velocity among elite and amateur male handball players. Int J Sports Med. 2005;26(3):225–32.
94. Ugrinowitsch C, Tricoli V, Rodacki AL, et al. Influence of training background on jumping height. J Strength Cond Res. 2007;21(3):848–52.
95. Drid P, Casals C, Mekic A, et al. Fitnss and anthropometric profiles of international vs. national judo medalists in half-heavyweight category. J Strength Cond Res. 2015;29(8):2115–21.
96. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power: part 2—training considerations for improving maximal power production. Sports Med. 2011;41(2):125–46.
97. Haff GG, Nimphius S. Training principles for power. Strength Cond J. 2012;34(6):2–12.
98. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power: part 1—biological basis of maximal power production. Sports Med. 2011;41(1):17–38.
99. Sole CJ. Analysis of countermovement vertical jump force-time curve phase characteristics in athletes [Doctoral Dissertation]. Digital Commons: East Tennessee State University; 2015.

100. Mizuguchi S. Net impulse and net impulse characteristics in vertical jumping: East Tennessee State University; 2012.
101. Cormie P, McGuigan MR, Newton RU. Changes in the eccentric phase contribute to improved stretch-shorten cycle performance after training. Med Sci Sports Exerc. 2010;42(9):1731–44.
102. Augustsson J, Thomee R. Ability of closed and open kinetic chain tests of muscular strength to assess functional performance. Scand J Med Sci Sports. 2000;10(3):164–8.
103. Comfort P, Stewart A, Bloom L, et al. Relationships between strength, sprint, and jump performance in well-trained youth soccer players. J Strength Cond Res. 2014;28(1):173–7.
104. Koch AJ, O'Bryant HS, Stone ME, et al. Effect of warm-up on the standing broad jump in trained and untrained men and women. J Strength Cond Res. 2003;17(4):710–4.
105. Loturco I, Nakamura FY, Artioli GG, et al. Strength and power qualities are highly associated with punching impact in elite amateur boxers. J Strength Cond Res. 2016;30(1):109–16.
106. McGuigan MR, Newton MJ, Winchester JB, et al. Relationship between isometric and dynamic strength in recreationally trained men. J Strength Cond Res. 2010;24(9):2570–3.
107. Secomb JL, Lundgren L, Farley ORL, et al. Relationships between lower-body muscle structure and lower-body strength, power and muscle-tendon complex stiffness. J Strength Cond Res. 2015;29(8):2221–8.
108. Secomb JL, Nimphius S, Farley ORL, et al. Relationships between lower-body muscle structure and lower-body strength, explosiveness and eccentric leg stiffness in adolescent athletes. J Sports Sci Med. 2015;14:691–7.
109. Ugarkovic D, Matavulj D, Kukolj M, et al. Standard anthropometric, body composition, and strength variables as predictors of jumping performance in elite junior athletes. J Strength Cond Res. 2002;16(2):227–30.
110. Wisløff U, Helgerud J, Hoff J. Strength and endurance of elite soccer players. Med Sci Sports Exerc. 1998;30(3):462–7.
111. Yamauchi J, Ishii N. Relations between force-velocity characteristics of the knee-hip extension movement and vertical jump performance. J Strength Cond Res. 2007;21(3):703–9.
112. Young WB, Wilson G, Byrne C. Relationship between strength qualities and performance in standing and run-up vertical jumps. J Sports Med Phys Fit. 1999;39(4):285–93.
113. Young WB, McLean B, Ardagna J. Relationship between strength qualities and sprinting performance. J Sports Med Phys Fit. 1995;35:13–9.
114. Bangsbo J, Nørregaard L, Thorsoe F. Activity profile of competition soccer. Can J Sport Sci. 1991;16(2):110–6.
115. Docherty D, Wenger HA, Neary P. Time-motion analysis related to the physiological demands of rugby. J Hum Mov Stud. 1988;14(6):269–77.
116. Reilly T. Energetics of high-intensity exercise (soccer) with particular reference to fatigue. J Sports Sci. 1997;15(3):257–63.
117. Gabbett TJ. Sprinting patterns of national rugby league competition. J Strength Cond Res. 2012;26(1):121–30.
118. Cunniffe B, Proctor W, Baker JS, et al. An evaluation of the physiological demands of elite rugby union using global positioning system tracking software. J Strength Cond Res. 2009;23(4):1195–203.
119. Duthie GM, Pyne DB, Marsh DJ, et al. Sprint patterns in rugby union players during competition. J Strength Cond Res. 2006;20(1):208–14.
120. Cometti G, Maffiuletti NA, Pousson M, et al. Isokinetic strength and anaerobic power of elite, subelite and amateur French soccer players. Int J Sports Med. 2001;22(1):45–51.
121. Chelly MS, Fathloun M, Cherif N, et al. Effects of a back squat training program on leg power, jump, and sprint performances in junior soccer players. J Strength Cond Res. 2009;23(8):2241–9.
122. Styles WJ, Matthews MJ, Comfort P. Effects of strength training on squat and sprint performances in soccer players. J Strength Cond Res. 2015. doi:10.1519/JSC.0000000000001243.
123. Comfort P, Haigh A, Matthews MJ. Are changes in maximal squat strength during preseason training reflected in changes in sprint performance in rugby league players? J Strength Cond Res. 2012;26(3):772–6.
124. Rønnestad BR, Nymark BS, Raastad T. Effects of in-season strength maintenance training frequency in professional soccer players. J Strength Cond Res. 2011;25(10):2653–60.
125. Rønnestad BR, Kvamme NH, Sunde A, et al. Short-term effects of strength and plyometric training on sprint and jump performance in professional soccer players. J Strength Cond Res. 2008;22(3):773–80.
126. Baker D, Nance S. The relation between running speed and measures of strength and power in professional rugby league players. J Strength Cond Res. 1999;13(3):230–5.
127. Chaouachi A, Brughelli M, Chamari K, et al. Lower limb maximal dynamic strength and agility determinants in elite basketball players. J Strength Cond Res. 2009;23(5):1570–7.
128. Harris NK, Cronin JB, Hopkins WG, et al. Relationship between sprint times and the strength/power outputs of a machine squat jump. J Strength Cond Res. 2008;22(3):691–8.
129. Lockie RG, Murphy AJ, Knight TJ, et al. Factors that differentiate acceleration ability in field sport athletes. J Strength Cond Res. 2011;25(10):2704–14.
130. McBride JM, Blow D, Kirby TJ, et al. Relationship between maximal squat strength and five, ten, and forty yard sprint times. J Strength Cond Res. 2009;23(6):1633–6.
131. Meckel Y, Atterbom H, Grodjinovsky A, et al. Physiological characteristics of female 100 metre sprinters of different performance levels. J Sports Med Phys Fitness. 1995;35(3):169–75.
132. Seitz LB, Trajano GS, Haff GG. The back squat and the power clean: elicitation of different degrees of potentiation. Int J Sports Physiol Perform. 2014;9(4):643–9.
133. Thomas C, Comfort P, Chiang C, et al. Relationship between isometric mid-thigh pull variables and sprint and change of direction performance in collegiate athletes. J Trainol. 2015;4(1):6–10.
134. Seitz LB, Reyes A, Tran TT, et al. Increases in lower-body strength transfer positively to sprint performance: a systematic review with meta-analysis. Sports Med. 2014;44(12):1693–702.
135. Barr MJ, Sheppard JM, Agar-Newman DJ, et al. Transfer effect of strength and power training to the sprinting kinematics of international rugby players. J Strength Cond Res. 2014;28(9):2585–96.
136. Spiteri T, Newton RU, Nimphius S. Neuromuscular strategies contributing to faster multidirectional agility performance. J Electromyogr Kinesiol. 2015;25(4):629–36.
137. Spiteri T, Newton RU, Binetti M, et al. Mechanical determinants of faster change of direction and agility performance in female basketball athletes. J Strength Cond Res. 2015;29(8):2205–14.
138. Young WB, Miller IR, Talpey SW. Physical qualities predict change-of-direction speed but not defensive agility in Australian rules football. J Strength Cond Res. 2015;29(1):206–12.
139. Spiteri T, Hart NH, Nimphius S. Offensive and defensive agility: a sex comparison of lower body kinematics and ground reaction forces. J Appl Biomech. 2014;30(4):514–20.
140. Barnes JL, Schilling BK, Falvo MJ, et al. Relationship of jumping and agility performance in female volleyball athletes. J Strength Cond Res. 2007;21(4):1192–6.
141. Marshall BM, Franklyn-Miller AD, King EA, et al. Biomechanical factors associated with time to complete a change of direction cutting maneuver. J Strength Cond Res. 2014;28(10):2845–51.

142. Alexander RM. Principles of animal locomotion. Princeton: Princeton University Press; 2003.
143. Weyand PG, Bundle MW, McGowan CP, et al. The fastest runner on artificial legs: different limbs, similar function? J Appl Physiol. 2009;107(3):903–11.
144. Spiteri T, Cochrane JL, Hart NH, et al. Effect of strength on plant foot kinetics and kinematics during a change of direction task. Eur J Sport Sci. 2013;13(6):646–52.
145. Hunter JP, Marshall RN, McNair PJ. Relationships between ground reaction force impulse and kinematics of sprint-running acceleration. J Appl Biomech. 2005;21(1):31–43.
146. Nimphius S, Geib G, Spiteri T, et al. "Change of direction deficit" measurement in division I American football players. J Aust Strength Cond. 2013;21(S2):115–7.
147. Nimphius S, Spiteri T, Seitz L, et al. Is there a pacing strategy during a 505 change of direction test in adolescents? [Abstract]. J Strength Cond Res. 2013;27(10):S104–5.
148. Sayers MGL. Influence of test distance on change of direction speed test results. J Strength Cond Res. 2015;29(9):2412–6.
149. Spiteri T, Nimphius S. Relationship between timing variables and plant foot kinetics during change of direction movements. J Aust Strength Cond. 2013;21(S1):73–7.
150. Spiteri T, Nimphius S, Hart NH, et al. Contribution of strength characteristics to change of direction and agility performance in female basketball athletes. J Strength Cond Res. 2014;28(9): 2415–23.
151. Delaney JA, Scott TJ, Ballard DA, et al. Contributing factors to change-of-direction ability in professional rugby league players. J Strength Cond Res. 2015;29(10):2688–96.
152. Jones P, Bampouras T, Marrin K. An investigation into the physical determinants of change of direction speed. J Sports Med Phys Fit. 2009;49(1):97–104.
153. Markovic G. Poor relationship between strength and power qualities and agility performance. J Sports Med Phys Fit. 2007;47(3):276–83.
154. Swinton PA, Lloyd R, Keogh JWL, et al. Regression models of sprint, vertical jump, and change of direction performance. J Strength Cond Res. 2014;28(7):1839–48.
155. Stone MH, Stone ME, Sands WA, et al. Maximum strength and strength training—a relationship to endurance? Strength Cond J. 2006;28(3):44–53.
156. Saunders PU, Pyne DB, Telford RD, et al. Factors affecting running economy in trained distance runners. Sports Med. 2004;34(7):465–85.
157. Aagaard P, Andersen JL, Bennekou M, et al. Effects of resistance training on endurance capacity and muscle fiber composition in young top-level cyclists. Scand J Med Sci Sports. 2011;21(6):e298–307.
158. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. Eur J Appl Physiol Occup Physiol. 1980;45(2–3):255–63.
159. Behm DG, Wahl MJ, Button DC, et al. Relationship between hockey skating speed and selected performance measures. J Strength Cond Res. 2005;19(2):326–31.
160. Dumke C, Pfaffenroth CM, McBride JM, et al. Relationship between muscle strength, power and stiffness and running economy in trained male runners. Int J Sports Physiol Perform. 2010;5:249–61.
161. Judge LW, Bellar D, Turk M, et al. Relationship of squat one repetition maximum to weight throw performance among elite and collegiate athletes. Int J Perform Anal Sport. 2011;11(2):209–19.
162. Judge LW, Bellar D. Variables associated with the personal best performance in the glide and spin shot put for US collegiate throwers. Int J Perform Anal Sport. 2012;12(1):37–51.
163. Reyes GF, Dolny D. Acute effects of various weighted bat warm-up protocols on bat velocity. J Strength Cond Res. 2009;23(7):2114–8.
164. Speranza MJA, Gabbett TJ, Johnston RD, et al. Relationship between a standardized tackling proficiency test and match-play tackle performance in semi-professional rugby league players. Int J Sports Physiol Perform. 2015;10(6):754–60.
165. Stone MH, Sanborn K, O'Bryant HS, et al. Maximum strength-power-performance relationships in collegiate throwers. J Strength Cond Res. 2003;17(4):739–45.
166. Stone MH, Sands WA, Pierce KC, et al. Relationship of maximum strength to weightlifting performance. Med Sci Sports Exerc. 2005;37(6):1037–43.
167. Tillin NA, Bishop D. Factors modulating post-activation potentiation and its effect on performance of subsequent explosive activities. Sports Med. 2009;39(2):147–66.
168. Suchomel TJ, Lamont HS, Moir GL. Understanding vertical jump potentiation: a deterministic model. Sports Med. 2015. doi:10.1007/s40279-015-0466-9.
169. Seitz LB, Haff GG. Factors modulating post-activation potentiation of jump, sprint, throw, and upper-body ballistic performances: a systematic review with meta-analysis. Sports Med. 2015. doi:10.1007/s40279-015-0415-7.
170. Miyamoto N, Wakahara T, Ema R, et al. Further potentiation of dynamic muscle strength after resistance training. Med Sci Sports Exerc. 2013;45(7):1323–30.
171. Stone MH, Sands WA, Pierce KC, et al. Power and power potentiation among strength-power athletes: preliminary study. Int J Sports Physiol Perform. 2008;3(1):55–67.
172. Jo E, Judelson DA, Brown LE, et al. Influence of recovery duration after a potentiating stimulus on muscular power in recreationally trained individuals. J Strength Cond Res. 2010;24(2):343–7.
173. Chiu LZF, Fry AC, Weiss LW, et al. Postactivation potentiation response in athletic and recreationally trained individuals. J Strength Cond Res. 2003;17(4):671–7.
174. Chiu LZF, Fry AC, Schilling BK, et al. Neuromuscular fatigue and potentiation following two successive high intensity resistance exercise sessions. Eur J Appl Physiol. 2004;92:385–92.
175. Bellar D, Judge LW, Turk M, et al. Efficacy of potentiation of performance through overweight implement throws on male and female collegiate and elite weight throwers. J Strength Cond Res. 2012;26(6):1469–74.
176. Bevan HR, Owen NJ, Cunningham DJ, et al. Complex training in professional rugby players: influence of recovery time on upper-body power output. J Strength Cond Res. 2009;23(6): 1780–5.
177. Chaouachi A, Poulos N, Abed F, et al. Volume, intensity, and timing of muscle power potentiation are variable. Appl Physiol Nutr Metab. 2011;36(5):736–47.
178. Duthie GM, Young WB, Aitken DA. The acute effects of heavy loads on jump squat performance: an evaluation of the complex and contrast methods of power development. J Strength Cond Res. 2002;16(4):530–8.
179. Judge LW, Bellar D, Craig B, et al. The influence of post activation potentiation on shot put performance of collegiate throwers. J Strength Cond Res. 2013. doi:10.1519/JSC.0b013e3182a74488.
180. Kilduff LP, Bevan HR, Kingsley MI, et al. Postactivation potentiation in professional rugby players: optimal recovery. J Strength Cond Res. 2007;21(4):1134–8.
181. Kilduff LP, Owen N, Bevan H, et al. Influence of recovery time on post-activation potentiation in professional rugby players. J Sports Sci. 2008;26(8):795–802.

182. Mangus BC, Takahashi M, Mercer JA, et al. Investigation of vertical jump performance after completing heavy squat exercises. J Strength Cond Res. 2006;20(3):597–600.
183. Okuno NM, Tricoli V, Silva SB, et al. Postactivation potentiation on repeated-sprint ability in elite handball players. J Strength Cond Res. 2013;27(3):662–8.
184. Ruben RM, Molinari MA, Bibbee CA, et al. The acute effects of an ascending squat protocol on performance during horizontal plyometric jumps. J Strength Cond Res. 2010;24(2):358–69.
185. Seitz LB, de Villarreal ESS, Haff GG. The temporal profile of postactivation potentiation is related to strength level. J Strength Cond Res. 2014;28:706–15.
186. Suchomel TJ, Sato K, DeWeese BH, et al. Potentiation effects of half-squats performed in a ballistic or non-ballistic manner. J Strength Cond Res. 2015. doi:10.1519/JSC.0000000000001251.
187. Suchomel TJ, Sato K, DeWeese BH, et al. Potentiation following ballistic and non-ballistic complexes: the effect of strength level. J Strength Cond Res. 2015. doi:10.1519/JSC.0000000000001288.
188. Terzis G, Spengos K, Karampatsos G, et al. Acute effect of drop jumping on throwing performance. J Strength Cond Res. 2009;23(9):2592–7.
189. Tsolakis C, Bogdanis GC. Influence of type of muscle contraction and gender on postactivation potentiation of upper and lower limb explosive performance in elite fencers. J Sports Sci Med. 2011;10(3):577–83.
190. West DJ, Cunningham DJ, Crewther BT, et al. Influence of ballistic bench press on upper body power output in professional rugby players. J Strength Cond Res. 2013;27(8):2282–7.
191. Witmer CA, Davis SE, Moir GL. The acute effects of back squats on vertical jump performance in men and women. J Sports Sci Med. 2010;9(2):206–13.
192. Young WB, Jenner A, Griffiths K. Acute enhancement of power performance from heavy load squats. J Strength Cond Res. 1998;12(2):82–4.
193. Berning JM, Adams KJ, DeBeliso M, et al. Effect of functional isometric squats on vertical jump in trained and untrained men. J Strength Cond Res. 2010;24(9):2285–9.
194. Gourgoulis V, Aggeloussis N, Kasimatis P, et al. Effect of a submaximal half-squats warm-up program on vertical jumping ability. J Strength Cond Res. 2003;17(2):342–4.
195. Rixon KP, Lamont HS, Bemben MG. Influence of type of muscle contraction, gender, and lifting experience on postactivation potentiation performance. J Strength Cond Res. 2007;21(2):500–5.
196. Batista MA, Roschel H, Barroso R, et al. Influence of strength training background on postactivation potentiation response. J Strength Cond Res. 2011;25(9):2496–502.
197. Jensen RL, Ebben WP. Kinetic analysis of complex training rest interval effect on vertical jump performance. J Strength Cond Res. 2003;17(2):345–9.
198. McBride JM, Nimphius S, Erickson TM. The acute effects of heavy-load squats and loaded countermovement jumps on sprint performance. J Strength Cond Res. 2005;19(4):893–7.
199. Bullock N, Comfort P. An investigation into the acute effects of depth jumps on maximal strength performance. J Strength Cond Res. 2011;25(11):3137–41.
200. Lehance C, Binet J, Bury T, et al. Muscular strength, functional performances and injury risk in professional and junior elite soccer players. Scand J Med Sci Sports. 2009;19(2):243–51.
201. Lehnhard RA, Lehnhard HR, Young R, et al. Monitoring injuries on a college soccer team: The effect of strength training. J Strength Cond Res. 1996;10(2):115–9.
202. Sole CJ, Kavanaugh AA, Reed JP, et al. The sport performance enhancement group: A five-year analysis of interdisciplinary athlete development. In: Beckham GK, Swisher A, editors. 8th Annual Coaches and Sport Science College. Johnson City; 2013. p. 28–30.
203. Jacobs C, Mattacola C. Sex differences in eccentric hip-abductor strength and knee-joint kinematics when landing from a jump. J Sport Rehabil. 2005;14(4):346–55.
204. Emery CA, Meeuwisse WH. The effectiveness of a neuromuscular prevention strategy to reduce injuries in youth soccer: a cluster-randomised controlled trial. Br J Sports Med. 2010;44(8):555–62.
205. Fleck SJ, Falkel JE. Value of resistance training for the reduction of sports injuries. Sports Med. 1986;3(1):61–8.
206. Kennedy MD, Fischer R, Fairbanks K, et al. Can pre-season fitness measures predict time to injury in varsity athletes? A retrospective case control study. BMC Sports Sci Med Rehab. 2012;4(1):26.
207. Lauersen JB, Bertelsen DM, Andersen LB. The effectiveness of exercise interventions to prevent sports injuries: a systematic review and meta-analysis of randomised controlled trials. Br J Sports Med. 2014;48(11):871–7.
208. Radin EL. Role of muscles in protecting athletes from injury. Acta Med Scand. 1986;220(S711):143–7.
209. Suchomel TJ, Bailey CA. Monitoring and managing fatigue in baseball players. Strength Cond J. 2014;36(6):39–45.
210. Sole CJ, Mizuguchi S, Suchomel TJ, et al. Longitudinal monitoring of countermovement jump mechanical variables: A preliminary investigation. In: Sato K, Sands WA, Mizuguchi S, editors. XXXIInd International Conference of Biomechanics in Sports. Johnson City; 2014. p. 159–62.
211. McMaster DT, Gill N, Cronin J, et al. A brief review of strength and ballistic assessment methodologies in sport. Sports Med. 2014;44(5):603–23.
212. Nimphius S. Lag time: the effect of a two week cessation from resistance training on force, velocity and power in elite softball players [Abstract]. J Strength Cond Res. 2010;24(Suppl 1).
213. Beckham GK, Lamont HS, Sato K, et al. Isometric strength of powerlifters in key positions of the conventional deadlift. J Trainol. 2012;1:32–5.
214. Bazyler CD, Sato K, Wassinger CA, et al. The efficacy of incorporating partial squats in maximal strength training. J Strength Cond Res. 2014;28(11):3024–32.
215. Sato K, Bazyler C, Beckham GK, et al. Force output comparison between six U.S. collegiate athletic teams. In: XXXth International Conference of Biomechanics in Sports 2012. Melbourne; 2012. p. 115–8.
216. Garhammer J. Power production by Olympic weightlifters. Med Sci Sports Exerc. 1980;12(1):54–60.
217. Haff GG, Stone M, O'Bryant HS, et al. Force-time dependent characteristics of dynamic and isometric muscle actions. J Strength Cond Res. 1997;11(4):269–72.
218. DeWeese BH, Sams ML, Serrano AJ. Sliding toward Sochi—part 1: a review of programming tactics used during the 2010–2014 quadrennial. Natl Strength Cond Assoc Coach. 2014;1(3):30–42.
219. Suchomel TJ, Sato K, DeWeese BH, et al. Relationships between potentiation effects following ballistic half-squats and bilateral symmetry. Int J Sports Physiol Perform. 2015. doi:10.1123/ijspp.2015-0321.
220. Baker D. The effects of an in-season of concurrent training on the maintenance of maximal strength and power in professional and college-aged rugby league football players. J Strength Cond Res. 2001;15(2):172–7.
221. Gorostiaga EM, Granados C, Ibáñez J, et al. Effects of an entire season on physical fitness changes in elite male handball players. Med Sci Sports Exerc. 2006;38(2):357–66.
222. Schmidt WD, Piencikowski CL, Vandervest RE. Effects of a competitive wrestling season on body composition, strength, and

power in National Collegiate Athletic Association Division III college wrestlers. J Strength Cond Res. 2005;19(3):505–8.
223. Hoffman JR, Fry AC, Howard R, et al. Strength, speed and endurance changes during the course of a division I basketball season. J Strength Cond Res. 1991;5(3):144–9.
224. Granados C, Izquierdo M, Ibanez J, et al. Effects of an entire season on physical fitness in elite female handball players. Med Sci Sports Exerc. 2008;40:351–61.
225. Stone MH, O'Bryant HS. Weight training: a scientific approach. Minneapolis: Burgess International; 1987.
226. Suchomel TJ, Beckham GK, Wright GA. Effect of various loads on the force-time characteristics of the hang high pull. J Strength Cond Res. 2015;29(5):1295–301.
227. Suchomel TJ, Comfort P, Stone MH. Weightlifting pulling derivatives: rationale for implementation and application. Sports Med. 2015;45(6):823–39.
228. Suchomel TJ, Taber CB, Wright GA. Jump shrug height and landing forces across various loads. Int J Sports Physiol Perform. 2015;11(1):61–5.
229. Suchomel TJ, Wright GA, Kernozek TW, et al. Kinetic comparison of the power development between power clean variations. J Strength Cond Res. 2014;28(2):350–60.
230. Suchomel TJ, DeWeese BH, Beckham GK, et al. The jump shrug: a progressive exercise into weightlifting derivatives. Strength Cond J. 2014;36(3):43–7.
231. Suchomel TJ, DeWeese BH, Beckham GK, et al. The hang high pull: a progressive exercise into weightlifting derivatives. Strength Cond J. 2014;36(6):79–83.
232. Suchomel TJ, Beckham GK, Wright GA. Lower body kinetics during the jump shrug: impact of load. J Trainol. 2013;2:19–22.
233. Comfort P, Udall R, Jones PA. The effect of loading on kinematic and kinetic variables during the midthigh clean pull. J Strength Cond Res. 2012;26(5):1208–14.
234. Comfort P, Allen M, Graham-Smith P. Kinetic comparisons during variations of the power clean. J Strength Cond Res. 2011;25(12):3269–73.
235. Comfort P, Allen M, Graham-Smith P. Comparisons of peak ground reaction force and rate of force development during variations of the power clean. J Strength Cond Res. 2011;25(5):1235–9.
236. Suchomel TJ, Wright GA, Lottig J. Lower extremity joint velocity comparisons during the hang power clean and jump shrug at various loads. In: Sato K, Sands WA, Mizuguchi S, editors. XXXIInd International Conference of Biomechanics in Sports. Johnson City; 2014. p. 749–52.
237. Young WB. Laboratory strength assessment of athletes. New Stud Athl. 1995;10:89–96.
238. Beckham GK, Suchomel TJ, Bailey CA, et al. The relationship of the reactive strength index-modified and measures of force development in the isometric mid-thigh pull. In: Sato K, Sands WA, Mizuguchi S, editors. XXXIInd International Conference of Biomechanics in Sports. Johnson City; 2014. p. 501–4.
239. Flanagan EP, Ebben WP, Jensen RL. Reliability of the reactive strength index and time to stabilization during depth jumps. J Strength Cond Res. 2008;22(5):1677–82.
240. Hamilton D. Drop jumps as an indicator of neuromuscular fatigue and recovery in elite youth soccer athletes following tournament match play. J Aust Strength Cond. 2009;17(4):3–8.
241. McClymont D. Use of the reactive strength index (RSI) as an indicator of plyometric training conditions. In: Reilly T, Cabri J, Araujo D, editors. Science and Football V: the Proceedings of the Fifth World Congress on Sports Science and Football. Lisbon: Routledge; 2003. p. 408–16.
242. Ebben WP, Petushek EJ. Using the reactive strength index modified to evaluate plyometric performance. J Strength Cond Res. 2010;24(8):1983–7.
243. Suchomel TJ, Bailey CA, Sole CJ, et al. Using reactive strength index-modified as an explosive performance measurement tool in Division I athletes. J Strength Cond Res. 2015;29(4):899–904.
244. Suchomel TJ, Sole CJ, Sams ML, et al. The effect of a competitive season on the explosive performance characteristics of collegiate male soccer players. In: Beckham GK, Swisher A, editors. 9th Annual Sport Science and Coaches College; 2014; Johnson City; 2014.
245. Suchomel TJ, Sole CJ, Bailey CA, et al. A comparison of reactive strength index-modified between six U.S. collegiate athletic teams. J Strength Cond Res. 2015;29(5):1310–6.
246. Bailey CA, Suchomel TJ, Beckham GK, et al. Reactive strength index-modified differences between baseball position players and pitchers. In: Sato K, Sands WA, Mizuguchi S, editors. XXXIInd International Conference of Biomechanics in Sports. 2014; Johnson City; 2014. p. 562–5.
247. Suchomel TJ, Sole CJ, Stone MH. Comparison of methods that assess lower body stretch-shortening cycle utilization. J Strength Cond Res. 2015. doi:10.1519/JSC.0000000000001100.
248. Keiner M, Sander A, Wirth K, et al. Strength performance in youth: trainability of adolescents and children in the back and front squats. J Strength Cond Res. 2013;27(2):357–62.
249. Ford P, De Ste Croix M, Lloyd R, et al. The long-term athlete development model: Physiological evidence and application. J Sports Sci. 2011;29(4):389–402.
250. Balyi I, Hamilton A. Long-term athlete development: trainability in childhood and adolescence. Olympic Coach. 2004;16(1):4–9.
251. Häkkinen K, Keskinen KL. Muscle cross-sectional area and voluntary force production characteristics in elite strength-and endurance-trained athletes and sprinters. Eur J Appl Physiol Occup Physiol. 1989;59(3):215–20.
252. Häkkinen K, Komi PV, Tesch PA. Effect of combined concentric and eccentric strength training and detraining on force-time, muscle fiber and metabolic characteristics of leg extensor muscles. Scand J Med Sci Sports. 1981;3:50–8.
253. Campos GE, Luecke TJ, Wendeln HK, et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. Eur J Appl Physiol. 2002;88(1–2):50–60.
254. Aagaard P, Andersen JL, Dyhre-Poulsen P, et al. A mechanism for increased contractile strength of human pennate muscle in response to strength training: changes in muscle architecture. J Physiol. 2001;534(2):613–23.
255. Kawakami Y, Abe T, Kuno S-Y, et al. Training-induced changes in muscle architecture and specific tension. Eur J Appl Physiol Occup Physiol. 1995;72(1–2):37–43.
256. Kawakami Y, Abe T, Fukunaga T. Muscle-fiber pennation angles are greater in hypertrophied than in normal muscles. J Appl Physiol. 1993;74(6):2740–4.
257. Duchateau J, Semmler JG, Enoka RM. Training adaptations in the behavior of human motor units. J Appl Physiol. 2006;101(6):1766–75.
258. Häkkinen K. Neuromuscular and hormonal adaptations during strength and power training. A review. J Sports Med Phys Fit. 1989;29(1):9.
259. Narici MV, Roi GS, Landoni L, et al. Changes in force, cross-sectional area and neural activation during strength training and detraining of the human quadriceps. Eur J Appl Physiol Occup Physiol. 1989;59(4):310–9.
260. Häkkinen K, Alen M, Komi PV. Changes in isometric force-and relaxation-time, electromyographic and muscle fibre characteristics of human skeletal muscle during strength training and detraining. Acta Physiol Scand. 1985;125(4):573–85.
261. Häkkinen K, Alen M, Kallinen M, et al. Neuromuscular adaptation during prolonged strength training, detraining and re-

strength-training in middle-aged and elderly people. Eur J Appl Physiol. 2000;83(1):51–62.
262. Häkkinen K, Newton RU, Gordon SE, et al. Changes in muscle morphology, electromyographic activity, and force production characteristics during progressive strength training in young and older men. J Gerontol A Biol Sci Med Sci. 1998;53(6):B415–23.
263. Behm DG. Neuromuscular implications and applications of resistance training. J Strength Cond Res. 1995;9(4):264–74.
264. Carolan B, Cafarelli E. Adaptations in coactivation after isometric resistance training. J Appl Physiol. 1992;73(3):911–7.
265. Rabita G, Pérot C, Lensel-Corbeil G. Differential effect of knee extension isometric training on the different muscles of the quadriceps femoris in humans. Eur J Appl Physiol. 2000;83(6):531–8.
266. Sale DG. Neural adaptation to resistance training. Med Sci Sports Exerc. 1988;20(5 Suppl):S135–45.
267. Sale DG. Neural adaptations to strength training. In: Komi PV, editor. Strength and power in sport. 2nd ed. Oxford: Blackwell Science; 2003. p. 281–313.
268. Semmler JG. Motor unit synchronization and neuromuscular performance. Exerc Sport Sci Rev. 2002;30(1):8–14.
269. Semmler JG, Kornatz KW, Dinenno DV, et al. Motor unit synchronisation is enhanced during slow lengthening contractions of a hand muscle. J Physiol. 2002;545(2):681–95.
270. Aagaard P, Simonsen EB, Andersen JL, et al. Neural inhibition during maximal eccentric and concentric quadriceps contraction: effects of resistance training. J Appl Physiol. 2000;89(6):2249–57.
271. Blazevich AJ, Cannavan D, Coleman DR, et al. Influence of concentric and eccentric resistance training on architectural adaptation in human quadriceps muscles. J Appl Physiol. 2007;103(5):1565–75.
272. Seynnes OR, de Boer M, Narici MV. Early skeletal muscle hypertrophy and architectural changes in response to high-intensity resistance training. J Appl Physiol. 2007;102(1):368–73.
273. Seynnes OR, Erskine RM, Maganaris CN, et al. Training-induced changes in structural and mechanical properties of the patellar tendon are related to muscle hypertrophy but not to strength gains. J Appl Physiol. 2009;107(2):523–30.
274. Kubo K, Ikebukuro T, Yata H, et al. Time course of changes in muscle and tendon properties during strength training and detraining. J Strength Cond Res. 2010;24(2):322–31.
275. Kubo K, Kawakami Y, Fukunaga T. Influence of elastic properties of tendon structures on jump performance in humans. J Appl Physiol. 1999;87(6):2090–6.
276. Bojsen-Møller J, Magnusson SP, Rasmussen LR, et al. Muscle performance during maximal isometric and dynamic contractions is influenced by the stiffness of the tendinous structures. J Appl Physiol. 2005;99(3):986–94.
277. Bompa TO, Haff G. Periodization: theory and methodology of training. Champaign: Human Kinetics; 2009.
278. Stone MH, Pierce KC, Sands WA, et al. Weightlifting: program design. Strength Cond J. 2006;28(2):10–7.
279. Kraemer WJ, Newton RU. Training for muscular power. Phys Med Rehabil Clin N Am. 2000;11(2):341–68.
280. Newton RU, Kraemer WJ. Developing explosive muscular power: implications for a mixed methods training strategy. Strength Cond J. 1994;16:20–31.
281. Mayhew JL, Ball TE, Bowen JC. Prediction of bench press lifting ability from submaximal repetitions before and after training. Sports Med Train Rehabil. 1992;3(3):195–201.